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Invited Review Article

# The artificial intelligence reformation of sustainable building design approach: A systematic review on building design optimization methods using surrogate models

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ABSTRACT

Artificial Intelligence (AI) applications in building performance prediction for environmental sustainability outcomes play a significant role in compensating for computationally incompetent and expensive approaches to solving increasingly complex optimization problems. Although machine-learning-based surrogate models (SMs), one of the many AI approximation strategies for higher-order models, have long been utilized in sophisticated optimization studies in various fields, their reliability in the field of sustainable and low-energy architectural design is now being extensively explored. To comprehend the effectiveness of applying state-of-the-art surrogate-assisted optimization methodologies for building energy performance, this study presents a comprehensive review of recent studies aimed at identifying innovative theories and critical factors for each component of the optimization process. Considering passive form and envelope design variables, seventy-two relevant studies from the Scopus database were selected for review after screening. The results, including the current trends for surrogate modeling and optimization methods, accuracy levels of SMs, extent of building performance improvement, and decisive building design variables, are thoroughly discussed according to the varying conditions of each study. Finally, a further discussion of recent methodological advancements and limitations can help identify potential approaches and challenges for future research.

## 1. Introduction

As a major contributor to global greenhouse gas (GHG) emissions and energy consumption [1,2], the building sector has been extensively investigated to mitigate its undesirable impacts on climate [3–5], natural resources [6,7], and human health [8–10]. Nonetheless, these impacts may be exacerbated by the projected expansion of the built environment until mid-century [11]. The anticipated two-fold increases in global urban land [11] and raw material consumption [12] are examples of unavoidable measures required to accommodate the incessant escalation of urban populations over the coming decades. Therefore, it is imperative to prioritize sustainable design strategies for both existing and new building stocks to develop resource-efficient, low-carbon, and climate change-resilient built environment [13].

Practical strategies for sustainable buildings have recently been influenced by advancements in performance evaluation techniques, computational capabilities, and communication technologies [14–16]. These advancements have been further accentuated by the involvement of Artificial Intelligence (AI) applications at various building scales and lifecycle stages [17–20]. Intelligent optimization and Machine Learning (ML), two of the most promising AI subfields [18], have consequently gained increasing attention in the fields of performance-oriented building design and retrofitting [21]. However, utilizing simulation-based building performance optimization (BPO) tools may require

*Abbreviations:* AI, Artificial Intelligence; ANN, Artificial Neural Networks; BP, Building Performance; BPO, Building Performance Optimization; CC, Climate Change; DL, Deep Learning; EUI, Energy Use Intensity; GA, Genetic Algorithm; GH, Grasshopper; GHG, Greenhouse Gas; GP, Gaussian Process; HVAC, Heating, Ventilation, and Air Conditioning; LCC, Lifecycle Cost; LCCE, Lifecycle Carbon Emissions; LHS, Latin Hypercube Sampling; MFNN, Multi-layer Feedforward Neural Network; ML, Machine Learning; MO, Multi-Objective; NSGA, Non-dominated Sorting GA; PSO, Particle Swarm Optimization; RBFN, Radial Basis Function Network; SA, Sensitivity Analysis; S-BPO, Surrogate-assisted BPO; sDA, spatial Daylight Autonomy; SM, Surrogate Model; SVM, Support Vector Machine; UDI, Useful Daylight Illuminance; WWR, Window-to-Wall Ratio.

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sophisticated human [22,23] and computational [24–26] resources to articulate complex problems, which has directed BPO practitioners to employ computationally inexpensive alternatives. These alternatives involved simplification methods, which proved their futile nature in yielding accurate performance feedback, and consequently, reliable optimal design suggestions [15,24,25].

Approximation methods, on the other hand, including surrogate (or meta-)modeling techniques, have long been effectively employed in engineering design optimization studies of various fields [27–32]. Surrogate models (SMs) emulate high-fidelity physics-based models by statistically correlating input (design variables) and output (performance targets) data from a sample within the design space [33]. While this underscores the predominance of prediction models based on actual building performance (BP) data, such as from sensors, in achieving realistic approximations, simulated (or synthetic) data remain advantageous for compensating technical and economic restrictions on actual data sufficiency [34]. Hence, a diverse range of surrogate modeling approaches based on simulated data have been explored in recent years, corroborating their suitability and significance for sustainable and low-energy building design practices [35,36]. This exploration highlights the potential superiority of ML-based methods, including Artificial Neural Networks (ANNs), for handling computationally intensive applications such as building design optimization [37,38].

Prior reviews of simulation-based optimization studies [14,15,24,25,39,40] have uncovered a set of promising concepts and guidelines for improved reliability, emphasizing the indispensability of "holistic optimizations" in attaining a meaningful spectrum of optimal solutions. In response to the increasing interest in optimizing passive design strategies [25], allowing the BPO problem to explore a broad range of design solutions can oppose the general perception of typical practitioners and offer substantial value to sustainable design processes [14,15]. In addition, considering the significant influence of the building's location and operational function in selecting the key optimization parameters [39], it is imperative then for holistic optimizations to accommodate various climates, building types, and lifecycle (LC) stages [40].

These recommendations further emphasize the potential advantages of surrogates-assisted BPO (S-BPO), particularly for complex or large-scale optimization problems with multiple objectives (MOs) [24,40]. However, the decisive process for determining the most suitable optimization approach is influenced by the unique effectiveness, computational efficiency, and implementation requirements of each BPO case. This highlights the advantage of local experimentation using various methods, including surrogate-assisted methods, instead of commonly used optimization algorithms and parameters [41,42]. Hence, consolidating the most effective S-BPO techniques is crucial to understand the adaptability of their concepts in future studies, particularly for optimizing sustainable passive design strategies.

### 1.1. This review

This study aims to explore the promising features of approximation applications in solving complex and seemingly impractical optimization problems, providing a better understanding of the extent to which AI contributes to the (re-)design of evidence-based energy-efficient, low-carbon, and low-cost buildings. This study provides a systematic review of the recent literature over the past ten years regarding advancements in sustainable architectural design optimization methods assisted by ML-based SMs. Additionally, this study focuses on the building form and envelope design variables as the main drivers of the optimization processes to further assess the effectiveness of passive design strategies as an initial step toward enhancing the environmental sustainability of buildings.

Referring to the individual behavior of sustainable design optimization practitioners, who may follow regional and temporal climatic, sociocultural, or economic variations, this study aims to synthesize diverse approaches from relevant work to uncover innovative theories, strengths, and limitations of the adopted methods. The findings of the current work in comprehending various methodologies are expected to support the development of standardized guidelines for applying S-BPO in future sustainable design research and practice. Hence, after discussing the selection criteria of the review sample in Section 2 and recent S-BPO trends and model performance insights in Section 3, this study presents a comprehensive overview, including passive design optimization applications (Section 4.1), influential design parameters (Section 4.2), generalization across diverse climates (Section 4.3), recent methodological advancements (Section 4.4), and the limitations and future possibilities of various S-BPO techniques (Section 4.5). Finally, Section 5 provides the concluding remarks and offers synoptic insights for future guidance and potential improvements.

## 2. Methodology

The review sample was selected through several stages of screening to include the most successful S-BPO strategies over the last decade adequately. The literature review began by searching the Scopus database using keywords related to the main targets of this study. Focusing on 1) the incorporation of SMs 2) that were trained and validated using simulated data 3) into building design optimization 4) to improve the environmental sustainability performance, selected keywords were accumulated to the initial search query to be "TITLE-ABS-KEY (building W/50 (energy OR sustainab* OR performance*) W/15 optimization* AND (surrogate OR meta* OR data-driven* OR (neural PRE/3 network) OR ((machine OR deep) PRE/3 *learning))) AND PUBYEAR>2013".

This query resulted in 1302 documents from various fields, which led to an automatic filtration process that excluded publications in which certain words were found in the title, abstract, or keywords. The exclusion keywords were selected based on the most common research fields among the results other than building energy or environmental performance optimization studies, such as machine vision, energy management or control, and structural optimization. The line "AND NOT ((energy W/15 manag*) OR ((system* OR predict*) PRE/10 (control* OR optimization)) OR chem* OR image OR structure)" was added to the query, which resulted in 550 papers that were proceeded, along with a selection of 31 papers from references, to the next stages of filtration (initial and thorough screening), as shown in the PRISMA flow diagram of Fig. 1.

The initial screening of 581 papers, based solely on a review of the title and abstract, led to the exclusion of 383 irrelevant studies spanning fields such as building system control and optimization, structural engineering, construction management, and computer engineering. Subsequently, the remaining 198 papers were subjected to a rigorous screening process, which resulted in the exclusion of additional 126 studies, as justified in Fig. 1. Ultimately, a total of seventy-two papers were considered for review after full-text screening, including seven sources that reported promising techniques for surrogate modeling applications in scopes other than the targeted S-BPO. Considerable efforts were made to dissect the various concepts and methodologies of each study in the selected review sample by verifying and categorizing the different concepts and strategies of each optimization phase. It is worth mentioning that so many techniques can be derived from the review sample, which directly affects the reliability of the entire methodology in generating optimal design solutions. However, processes irrelevant to the main target, such as the simulation model setup and calibration, or case-specific configurations and design constraints, were eliminated from the analysis. Finally, the consolidated analyses in Table A1 are divided into four main groups: 1) case study configuration, 2) surrogate modeling strategies, 3) BPO strategies, and 4) selected design variables (inputs) and objectives (outputs).

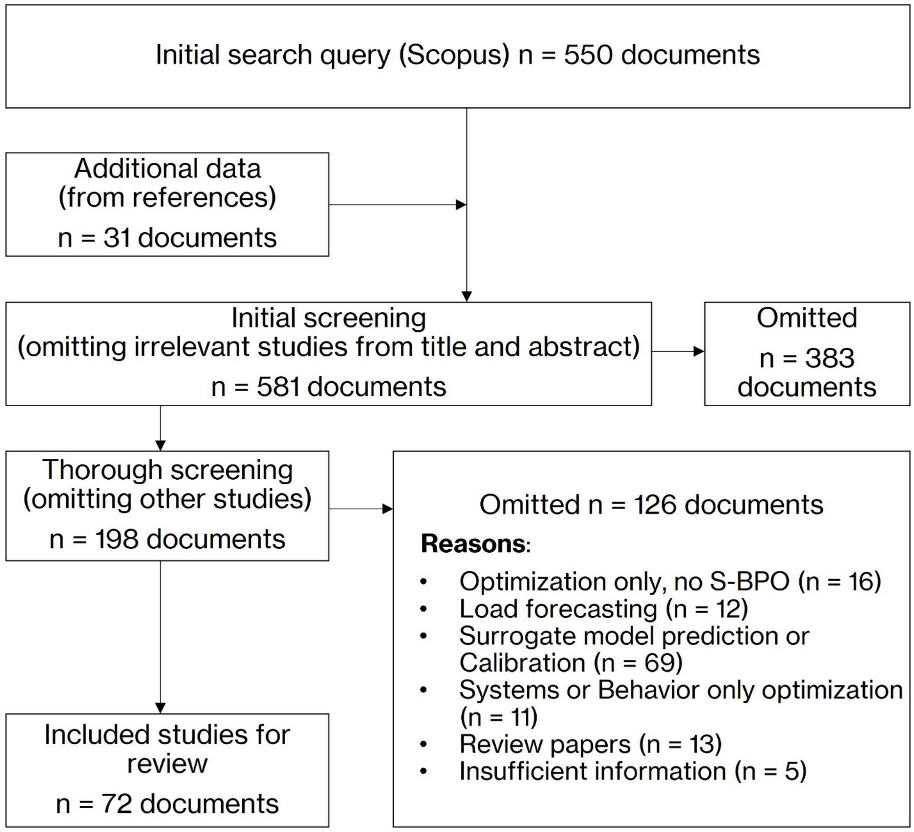


**Fig. 1.** PRISMA flowchart diagram of the excluded and included articles for review.

## 3. Results

Tracing the publication frequency over the last two decades, as plotted in Fig 2, shows that ML-based S-BPO, which mainly considers passive design parameters, has recently gained much attention among architectural design optimization researchers. In particular, the last five years have witnessed an exponential increase in the number of studies, implying a habitual interrelationship between BPO and SMs. The methodologies for performance prediction and optimization, their estimation accuracies, and their contributions to the energy and environmental performance improvement of buildings are presented in the following sections.

### 3.1. Current trends for S-BPO applications

An overview of the case study variations is presented in Table A1 and Fig. 3. Building type variations (Fig. 3/a) shows residential buildings as the most frequent type of investigation (41 %, 29 publications), ranging from single detached houses to high-rise apartments. Offices and commercial buildings were the second most studied types (19 publications), followed by educational buildings (12 publications). Public buildings constituted 10 % of the review sample, including libraries, tourist centers, and sports halls. No building type was specified in three publications, as it was found to have a negligible impact on optimization. For example [43], optimized wind amplification in a duct through a high-rise building for renewable energy generation. In addition [44], enhanced the annual total heat gain and heat flux through an uncertainty-based optimization of the thermophysical properties of a

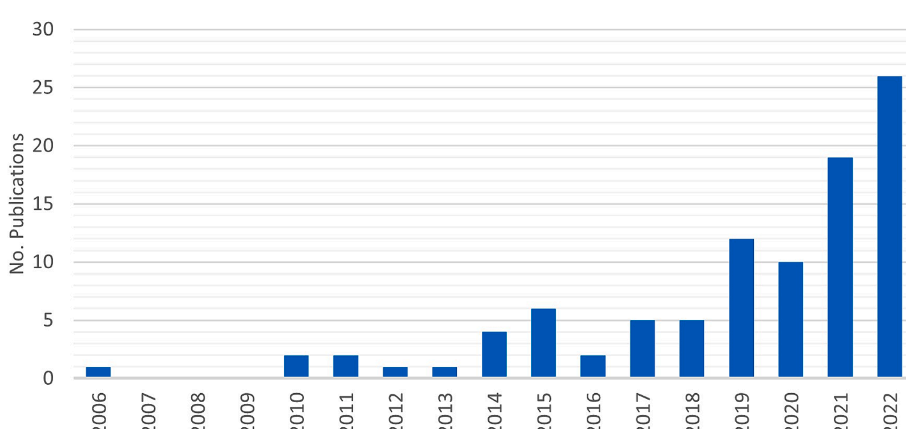


**Fig. 2.** Number of studies that involved S-BPO for building form and envelope design parameters per year.

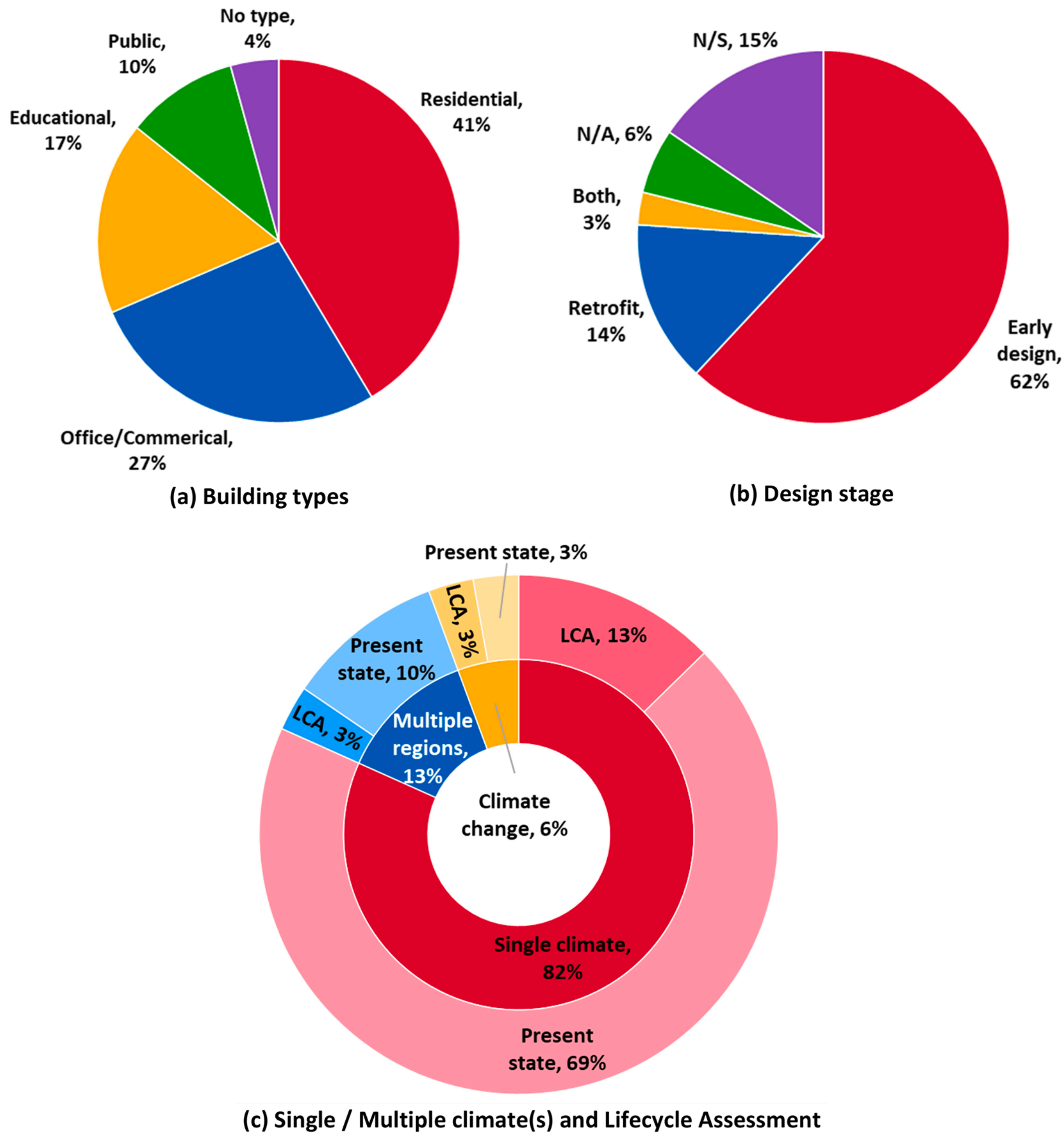


**Fig. 3.** Case study variations from seventy-two publications according to different (a) building types, (b) building design stages, (c) considerations for Lifecycle Assessment (LCA) optimization within single or multiple climates. Note: N/S=Not specified, and N/A=Not Applicable.

climate-adaptive aerogel glazing system. Moreover, within the four categories, the publications demonstrated disparities in design stages, building sizes, and climates, leading to the utilization of widely diverse methodologies for review.

The breakdown in Fig. 3(b) indicates that forty-four studies considered the early design stage for optimization, ten studies focused on retrofitting, two studies reported expedient methodologies for both stages, and eleven studies did not specify the design stage. Additionally, four publications were not applicable for design-stage categorization, which optimized operational amendments such as occupant behavior [45], furniture layout [46], and shading angles [47]. Further classifications in Fig. 3(c) show that nine studies optimized the same building model in multiple climatic regions, either locally or in different countries. In addition, four studies explored the impact of climate change (CC) on the performance prediction of buildings over their lifespan, thereby offering an opportunity to consider the life cycle environmental impact or cost optimization objectives, as in [48,49]. However, of the nine studies that considered a Life Cycle Assessment (LCA), seven overlooked the effect of CC on the performance of their optimal design solutions. Furthermore, the analyzed locations of the reviewed S-BPO studies encompassed 14 distinct global climate zones (Table A1), as determined by the Köppen-Geiger global classification of the present-day climate [50]. This classification is further employed in the present study to support the enriched discussion in Section 4.

### 3.1.1. *Optimization design parameters and objectives*

Categorizing the design parameters reveals that the fenestration geometry, including glazing area or shading projection, is the predominant group of variables (72 % of studies). This was followed by solid envelope and fenestration material parameters, encompassing thermo-physical, radiational, and optical properties (65 % and 61 %, respectively), along with building shape and geometry parameters (61 %). Besides, energy related indicators apparently dominated the performance objectives (77 %) due to the initial search query in Section 2, followed by thermal comfort (56 %), daylighting and visual comfort (25 %), general and energy costs (24 %), and carbon emissions (17 %). These percentages exceed 100 % due to combining objectives or groups of variables in the same study. As shown in the histograms in Fig. 4, 65 % of studies considered four or more unique design parameters, whereas 83 % conducted multi-objective (MO) BPO. A comprehensive list of 21 distinct design parameters for each study is available in Supplementary Table S1. However, the actual variable dimensionality is often higher

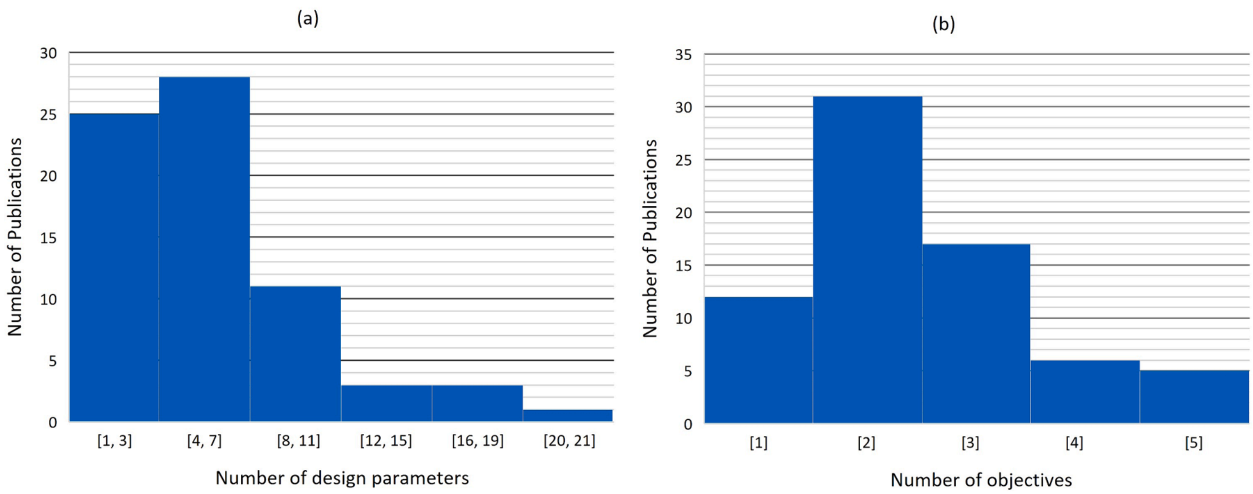


**Fig. 4.** Histogram of (a) the numbers of parameter types and (b) number of objectives according to the number of studies.

| | | Renewable Energy Yield | Energy Demand / Consumption | Energy Cost | Carbon Emissions | Thermal Comfort | Dynamic Daylight | Visual Comfort | General Cost | Others |
|---|---|---|---|---|---|---|---|---|---|---|
| Urban context | Depth of peripheral zone | 0% | 4% | 0% | 0% | 1% | 0% | 0% | 0% | 0% |
| | Outdoor airflow | 0% | 1% | 0% | 1% | 1% | 0% | 0% | 0% | 0% |
| | Building height | 1% | 1% | 0% | 0% | 3% | 3% | 1% | 0% | 3% |
| | Aspect ratio (H/W) / Compactness | 1% | 3% | 0% | 0% | 1% | 1% | 0% | 0% | 1% |
| | Open space ratio / SVF | 1% | 3% | 0% | 0% | 0% | 1% | 0% | 0% | 0% |
| | Solar air temperature | 0% | 3% | 0% | 0% | 4% | 0% | 0% | 0% | 0% |
| Shape /Geometry | Space area / dimensions | 3% | 20% | 1% | 1% | 13% | 7% | 3% | 0% | 4% |
| | Shape coefficient | 1% | 4% | 0% | 0% | 3% | 1% | 0% | 0% | 0% |
| | Floor numbers | 3% | 10% | 1% | 1% | 4% | 3% | 3% | 0% | 1% |
| | Floor height | 1% | 8% | 1% | 0% | 7% | 7% | 0% | 1% | 3% |
| | Orientation | 3% | 30% | 1% | 6% | 20% | 10% | 1% | 7% | 7% |
| | Overhang projection / Shading factor | 0% | 7% | 0% | 3% | 10% | 1% | 0% | 4% | 0% |
| | Atriums / duct openings | 1% | 1% | 0% | 0% | 0% | 1% | 0% | 0% | 1% |
| | Furniture layout | 0% | 1% | 0% | 0% | 0% | 0% | 0% | 0% | 0% |
| Envelope Materials | Thermal transmittance / resistance (walls) | 1% | 21% | 3% | 8% | 14% | 0% | 0% | 8% | 0% |
| | Thermal transmittance / resistance (roofs) | 0% | 15% | 1% | 7% | 13% | 0% | 0% | 8% | 0% |
| | Thermal transmittance / resistance (floors) | 0% | 7% | 0% | 0% | 4% | 0% | 0% | 0% | 0% |
| | Thermal absorptance | 0% | 1% | 0% | 3% | 4% | 0% | 0% | 3% | 0% |
| | Thermal conductivity | 1% | 8% | 1% | 0% | 10% | 4% | 0% | 0% | 1% |
| | Visible absorptance | 0% | 1% | 0% | 3% | 4% | 1% | 0% | 3% | 0% |
| | Solar absorbtance / SHGC | 1% | 21% | 1% | 4% | 18% | 6% | 1% | 4% | 0% |
| | Insulation thickness | 1% | 21% | 1% | 4% | 15% | 3% | 1% | 7% | 3% |
| | Insulation type | 0% | 11% | 1% | 6% | 6% | 0% | 1% | 4% | 1% |
| | Insulation area | 0% | 1% | 0% | 1% | 0% | 0% | 0% | 0% | 0% |
| | Wall / roof / floor thickness | 1% | 14% | 1% | 1% | 14% | 6% | 3% | 0% | 3% |
| | Construction type | 0% | 4% | 0% | 0% | 1% | 1% | 0% | 3% | 0% |
| | Density | 1% | 8% | 1% | 3% | 13% | 6% | 0% | 3% | 1% |
| | Specific heat | 1% | 11% | 1% | 0% | 10% | 6% | 0% | 0% | 1% |
| | Thermal transmittance (thermal bridge) | 0% | 4% | 0% | 0% | 1% | 0% | 0% | 0% | 0% |
| Fenestration Geometry | Window / glazing area ratio | 1% | 45% | 1% | 10% | 32% | 10% | 3% | 10% | 4% |
| | Window height | 0% | 3% | 0% | 0% | 1% | 1% | 0% | 0% | 0% |
| | Window disribution | 0% | 1% | 0% | 0% | 0% | 0% | 0% | 0% | 0% |
| | Glazing orientation | 0% | 1% | 0% | 0% | 3% | 0% | 0% | 0% | 1% |
| | Shading projection / Spacing | 1% | 24% | 1% | 1% | 15% | 11% | 3% | 3% | 1% |
| | Shading agle | 0% | 6% | 0% | 0% | 6% | 6% | 1% | 0% | 1% |
| | Air change rate | 1% | 25% | 3% | 8% | 18% | 1% | 1% | 10% | 1% |
| | Ventilation rate / open status | 0% | 7% | 0% | 0% | 7% | 0% | 1% | 0% | 0% |
| Fenestration Materials | Thermal transmittance | 0% | 23% | 0% | 6% | 18% | 4% | 0% | 8% | 1% |
| | Solar absorbtance / SHGC | 0% | 20% | 0% | 6% | 15% | 4% | 0% | 7% | 0% |
| | Visible light transmittance | 0% | 7% | 0% | 4% | 8% | 4% | 0% | 3% | 0% |
| | Reflectance | 0% | 1% | 0% | 0% | 1% | 0% | 0% | 0% | 0% |
| | Solar transmittance | 0% | 4% | 0% | 1% | 4% | 0% | 0% | 1% | 1% |
| | Glazing / frame type | 1% | 25% | 3% | 6% | 18% | 6% | 3% | 8% | 1% |
| HVAC /Systems | Heating setpoint | 1% | 18% | 3% | 4% | 18% | 3% | 3% | 7% | 0% |
| | Cooling setpoint | 1% | 20% | 1% | 6% | 20% | 3% | 3% | 7% | 0% |
| | De/humidification setpoint | 0% | 3% | 1% | 0% | 3% | 0% | 0% | 1% | 0% |
| | Supply air (/ water) temperature | 0% | 6% | 0% | 3% | 8% | 0% | 0% | 3% | 1% |
| | Air (/ water) intensity | 1% | 6% | 1% | 4% | 8% | 0% | 1% | 3% | 1% |
| | Time | 0% | 3% | 1% | 0% | 1% | 0% | 0% | 3% | 0% |
| | Type / heating or cooling method | 1% | 7% | 4% | 3% | 4% | 0% | 0% | 6% | 0% |
| | Efficiency / power | 1% | 13% | 1% | 7% | 11% | 0% | 1% | 6% | 3% |
| Renewables | PV / solar collectors area | 1% | 6% | 1% | 3% | 4% | 0% | 0% | 3% | 1% |
| | Slope | 0% | 1% | 0% | 1% | 1% | 0% | 0% | 1% | 1% |
| | storage volume | 0% | 1% | 0% | 1% | 1% | 0% | 0% | 1% | 1% |
| | PV / wind turine / biomass power | 0% | 3% | 0% | 1% | 1% | 0% | 0% | 1% | 0% |
| | Type | 0% | 1% | 0% | 0% | 1% | 0% | 0% | 1% | 0% |
| appliances /lights | Intensity / status | 0% | 4% | 0% | 4% | 7% | 0% | 3% | 3% | 1% |
| | Equipment gains | 0% | 1% | 0% | 1% | 1% | 0% | 0% | 0% | 0% |
| | Type | 0% | 6% | 1% | 0% | 4% | 0% | 0% | 4% | 0% |

0% 45%

**Fig. 5.** Co-occurrence of each of design variables and optimization objectives in relation to the number of studies.

because one parameter type, for example, envelope thermal characteristics or glazing area, can be applied to multiple façade orientations.

Similarly, to better understand the co-occurrence of each design parameter and performance objective, their frequencies according to the number of reviewed studies are mapped in detail in Fig. 5. The most frequent parameter for almost all objectives was the window area ratio, such as the window-to-wall ratio (WWR) or window-to-floor ratio (WFR), followed by building orientation, glazing type, air change rate, and shading device dimensions for energy performance objectives. Moreover, for thermal comfort objectives, the most frequent parameters, following the window area ratio, were building orientations, heating and cooling setpoints, and glazing materials. It is noteworthy that the percentages in Fig. 5 are not necessarily specific to the same objective because most studies have conducted MO optimization. Therefore, the design parameters may have been incorporated into the same optimization method to solve another objective.

### 3.1.2. Building performance data gathering and sampling strategy

A presumptively unsophisticated variable dimensionality can lead to millions of unique solutions for evaluation [51], emphasizing the significance of an efficient sampling strategy that would reduce computational costs and develop BPO-effective SMs [52]. The sampling strategy comprised four essential components extracted from each study:1) sampling type, 2) sampling method, 3) sample size, and 4) sample division (Table A1).

The sampling types were classified as static and adaptive (or active). Whereas both types randomly select representative samples from the design space, adaptive sampling actively updates the SM, particularly in promising areas of the search space based on optimization objectives, for example, low energy consumption. Only 11 % of the reviewed studies employed an adaptive sampling strategy despite its advantages of improved optimization convergence and accuracy, as further discussed in Section 4.4.2. In fact [52], underlined the inefficiency of static sampling from the perspective of S-BPO because there is no need to randomly scatter samples in non-interesting areas. Nevertheless, an initial sample of sufficient size is necessary for adaptive sampling studies to commence the optimization process, following the same criteria as a static sampling technique.

The frequency of the adopted sampling methods, presented in Fig. 6, shows that the most used method (40 %) was Latin Hypercube Sampling (LHS), followed by Random Sampling (9 %). In addition, three studies

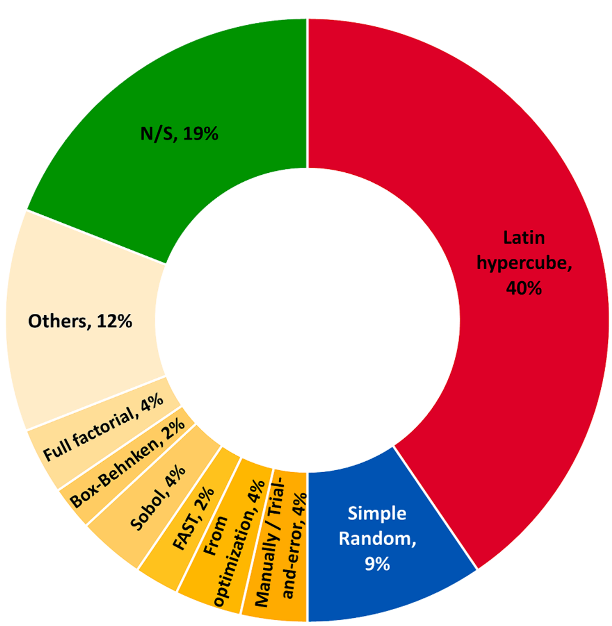


**Fig. 6.** Frequency of the adopted sampling methods, N/S means non-specified.

generated samples from an optimization study to train the SM for another type of prediction, as in [53] which developed an SM from 5000 genetic populations to classify (label) design schemes based on their performance. Regarding sample size and division, Fig. 7(a) shows that the number of samples ranged from 15 to 600,000 (off-chart), with a median of less than 800 sample points for all studies. After collection and preprocessing, e.g., normalizing the data, it was usually divided into three parts: training, validation, and testing, with median percentages, as shown in Fig. 7(b), of 80 %, 15 %, and 20 %, respectively. The review also indicated the lack of a uniform approach to correlate sample size with the number of inputs. However, the selection of the sample size and sampling method significantly influences the model accuracy, as indicated in Section 3.2. In addition, most studies used synthetic BP data for performance feedback, suggesting a cautious selection of the sample size to moderate the computational burden.

### 3.1.3. Training and tuning surrogate models

The most frequent SM types, as shown in Fig. 8(a), are those based on ANNs, with the dominance of Multi-Layer Feedforward Neural Networks (MFNN) (37 %), followed by Radial Basis Function Networks (RBFN) (5 %). Regression models were found to be in the second place with a combined percentage of 14 %, while Support Vector Machine (SVM) and Gaussian Process (GP) were the third common types, with percentages of 7 % each. Despite their growing interest in BP prediction, Deep Learning (DL) models were explicitly utilized in only 4 S-BPO studies, following the typical structure of Multi-Layer Feedforward Deep Neural Networks (MFDNNs). However, 49 % of ANN-based models featured a similar structure of multiple hidden layers, which are likely to enhance generalization through consecutive levels of feature abstraction. Potential advantages and limitations of DL models, as well as the strategies for incorporating advanced forms of Deep Neural Networks (DNNs) in S-BPO, are further discussed in Section 4.4.3.

In addition to single-learning algorithms, 12 studies employed ensemble learning techniques to improve prediction performance, as discussed in Section4.4.4. The composite model nullifies the limitations of the separately developed base models by combining their final outcomes. However, the overall accuracy of the ensembles of various model types is determined by the performance of each SM, which should first be properly trained to minimize errors [54].

Regarding the tuning process shown in Fig. 8(b), that is, the hyperparameters and structure selection and optimization, most studies manually selected these parameters by implementing a trial-and-error method. While this method was considered laborious and time-intensive, it was found to be efficient in developing accurate models, as discussed in Section 3.2. Alternatively, studies have revealed other types of SM tuning such as grid and random searches. The former searches for all possible combinations of hyperparameters (i.e., an exhaustive search method), whereas the latter selects only random samples to evaluate the model performance, which is time efficient. Other than that, 13 % of studies incorporated an optimization algorithm, such as Genetic Algorithm (GA), e.g., in [55,56], or Particle Swarm Optimization (PSO) algorithm, e.g., in [57], in which the optimization function is to minimize the prediction errors. Further implications of the hyperparameter optimization using hybrid methods for computationally efficient modeling are discussed in section 4.4.4.

### 3.1.4. Optimization algorithms

Once the inputs and outputs of an SM are successfully correlated, it can replace several simulation-based applications in the same climatic context [51], including combinations with a design search algorithm to satisfy certain performance criteria. As shown in Fig. 9, GAs constitute nearly two-thirds of all the studies, including Multi-Objective GA (MOGA), Non-dominated Sorting GA-II (NSGA-II), and NSGA-III. The second most common family of algorithms was PSO, which was employed in ten studies, including Multi-Objective PSO (MOPSO), PSO algorithm with Inertia Weight (PSOIW), Optimized MOPSO (OMOPSO),

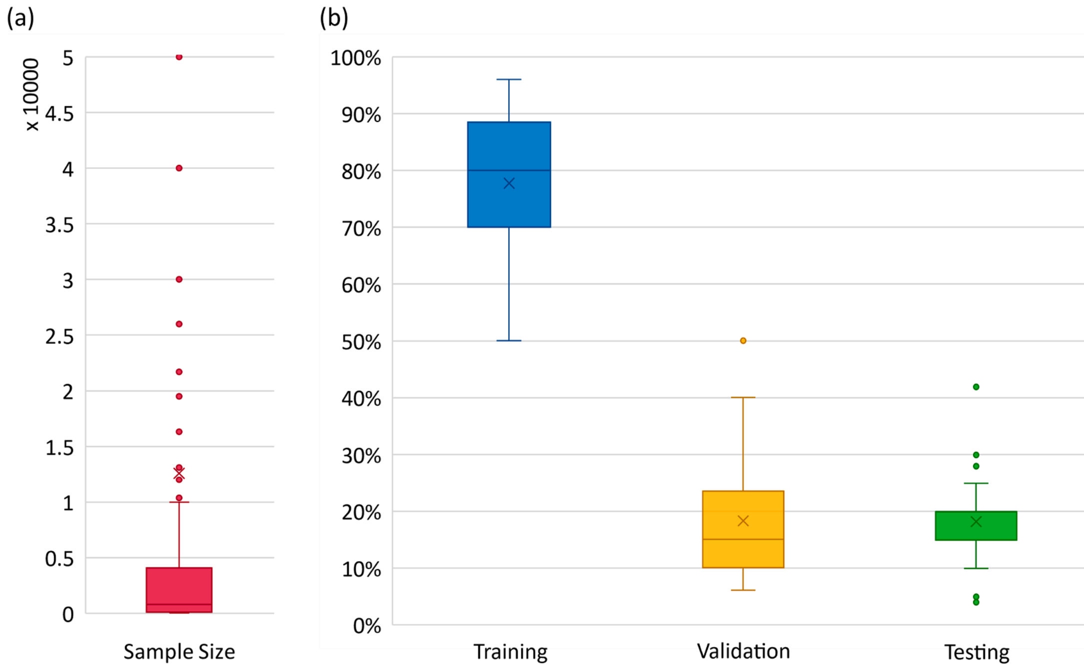


**Fig. 7.** (a) Range of sample sizes for all studies, and (b) the sample division percentage for training, validation, and testing SMs.

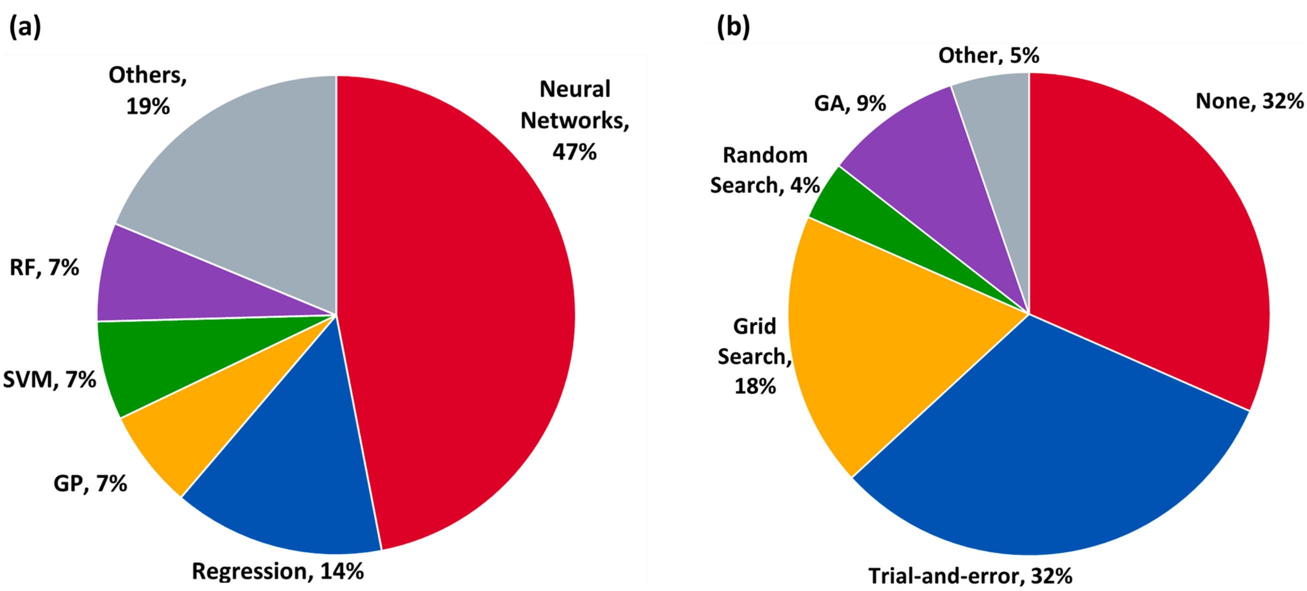


**Fig. 8.** Frequency of (a) SM types, and (b) SM tuning methods.

and Speed-constrained Multi-objective PSO (SMPSO). Overall, eighty-six algorithms were employed in the seventy-two studies, where eleven studies compared the convergence speed and quality of the optimal solutions derived using two or more algorithms.

### 3.2. Estimation accuracy of surrogate models

The approximation accuracy of the reviewed SMs usually depends on factors such as the model complexity, hyperparameter tuning, and sampling strategy [58]. Approximately 21 % of studies assessed the effect of various sample sizes and methods on model accuracy to determine the recommended sampling strategy for S-BPO. For instance, Ref. [59] compared the error values for ANN-based SMs with five datasets ranging from 100 to 3000 sample points and suggested a moderate sample size of 1,000 for acceptable accuracy. Alternatively, Ref. [60] compared the same sample size from five different methods, which interestingly favored the Sobol method in representing the entire search space. This choice highlights the preponderance of case-specific methods and configurations rather than frequently used ones.

Similarly, it is evident that there is no clear method for generalizing other accuracy-influencing factors to all models. To encapsulate the information from the reviewed studies, the parallel plot in Fig. 10 correlates the accuracy of the 176 SMs, represented by the coefficient of determination ($R^2$), with the number of inputs, tuning method, number of outputs per model, and type of outputs (performance indicators). The size of each bar in all columns denotes the percentage of models with the same characteristics, whereas the colors indicate $R^2$ ranges. The nonuniform $R^2$ scale was determined based on the shared coefficients among the models. In addition, each curve provides insight into the adopted methodology for any model that reaches a specific $R^2$ value.

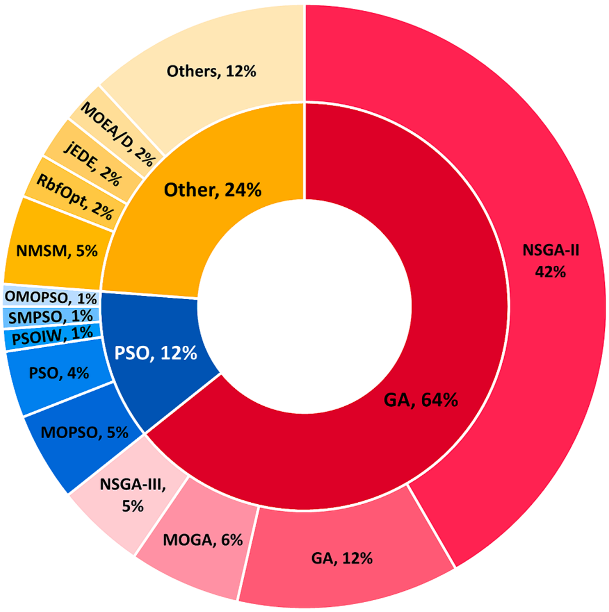


**Fig. 9.** Frequency of optimization algorithms. *Note*: NMSM=Nelder–Mead Simplex Method (with desirability function); jEDE=Self-adaptive Differential Evolution algorithm with an ensemble of mutation strategies; MOEA/D=MO Evolutionary Algorithm (EA) based on decomposition.

Other factors, such as adaptive sampling or model type, were excluded because of the limited number of studies covering various aspects of the same factor. For example, model type comparison was avoided because of the predominance of ANN-based models (137 out of 176 models), which typically constitute nearly 86 % of SMs, with $R^2 > 0.99$. Although the median $R^2$ values for most model types exceed 0.98, these accuracy values cannot be generalized to future S-BPO methods.

Most studies have affirmed that the higher the model dimensionality represented in the number and ranges of inputs, the lower the accuracy that can be achieved. In response, all models with more than 20 inputs (67 models) were based on ANNs, 84 % of which exploited a deep structure using multiple hidden layers. However, nearly half of these DNN models could not realize an $R^2$ value greater than 0.95. In general, nearly 79 % of ANN-based models that did not achieve an $R^2$ value greater than 0.9 included more than 20 inputs, whereas the remaining models comprised only six inputs. This indicates that the accuracy trend related to dimensionality is not perpetual, as the lowest accuracies ($R^2 < 0.76$) can occur even for optimized SMs with eleven or fewer inputs. Moreover, it is evident from Fig. 10 that the generally approved SM tuning settings can achieve high prediction accuracies for thermal comfort and energy consumption, even in complex studies with 19 inputs. Most of these models achieved $R^2 > 0.995$, where the model structures and hyperparameters were selected based on a literature survey or researchers' past experiences.

Regarding the model outputs, despite a substantial number of models that predicted dynamic daylight, such as spatial Daylight Autonomy (sDA) and Useful Daylight Illuminance (UDI), only one model achieved an $R^2$ of less than 0.850. The potentially higher prediction accuracy of daylight models is justified, as in [61], because of the comparably lower number of design variables with the same training sample as for the other energy- or thermal-related metrics. Similarly, a tendency to develop two or more SMs, each predicting a single performance indicator, was identified in MO-BPO studies to maintain model accuracy. Developing different SMs can be beneficial for exhibiting several hyperparameters, structures, or even model types, as in [45,62], according to the performance indicator requirements. It is noteworthy that, even though tuned ANN-based SMs in Ref. [62] achieved the highest performance, SVM with slightly lower accuracy was selected for the optimization process of thermal comfort due to its simplicity and operational efficiency. Anyway, studies that relate SM development to various output types can provide more reliable comparisons of the prediction accuracy owing to the presence of coequal circumstances. Ref. [63] reported lower accuracy levels for predicting Annual Sunlight Exposure (ASE) compared to sDA after developing 20 SMs that varied according to the design scenarios, zones, and model output. In addition [64], clarified that the increased complexity of large-scale structural

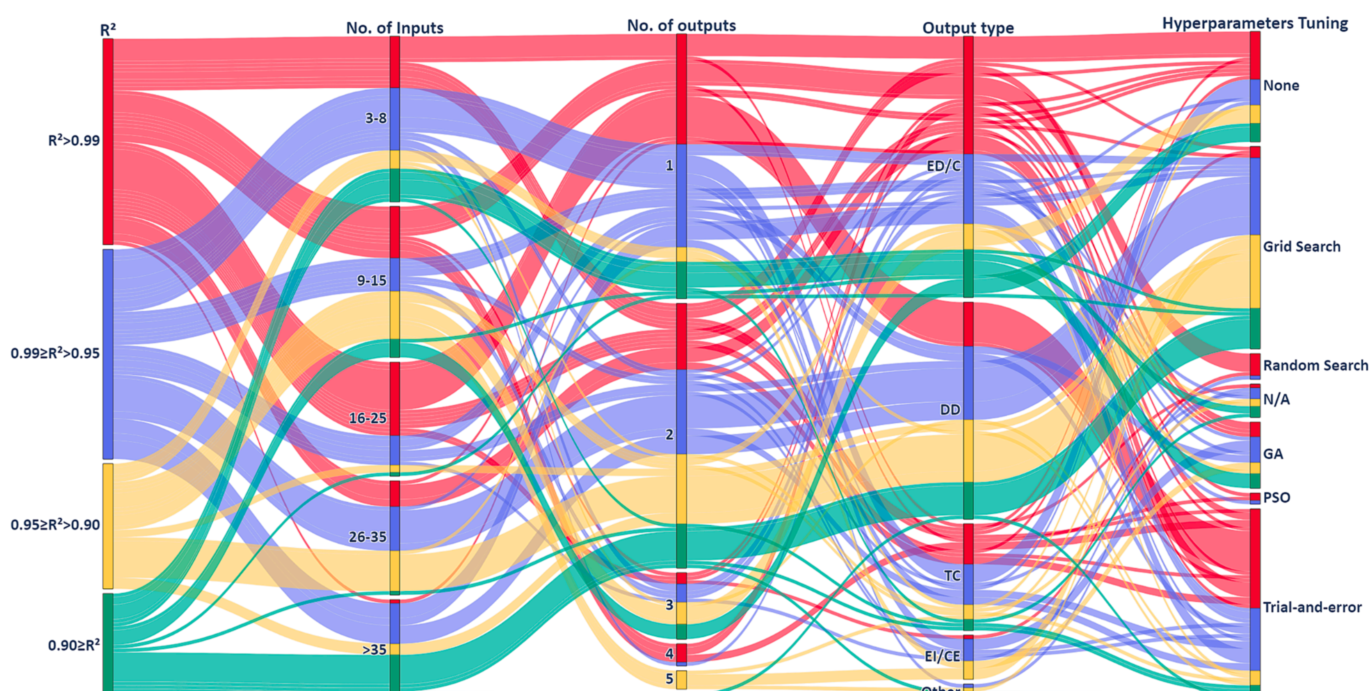


**Fig. 10.** Categorical parallel plot for $R^2$ values of 176 SMs and four of the possible influential factors. ED/C=Energy demand or consumption, DD=Dynamic daylight, TC=Thermal comfort, EI/ CE=Environmental Impacts or Carbon Emissions.

SMs yields lower accuracy levels compared to energy performance. However, the developed SM demonstrated sufficient effectiveness in supporting geometric decision making within a feasible structural range [65].

Further, SMs with exponential accuracy levels do not necessarily perform equally well during optimization. Therefore, from the perspective of the BPO, the accuracy of the optimization results, that is, the iterations near the optimal solutions, should also be evaluated. In MOO, a set of optimal solutions, such as the Pareto front of non-dominated solutions, is generated, representing trade-offs between the local optima of each objective. To assess the quality of these solutions, the investigation can involve selecting a subset from the S-BPO optimal solutions (approximated optimal solutions) for a simulation-based validation process, as in Ref. [51]. While this approach can be resource efficient, other studies have opted for a complete simulation-based optimization for more rigorous comparisons. In these cases, the entire set of approximated optimal solutions can be evaluated by measuring their closeness to either the optimal solution set obtained through an exhaustive search method (True optimal solutions), as in [35,66], or through intelligent optimization algorithms (Reference optimal solutions), as in [67,68].

## 4. Discussion

### 4.1. General remarks on surrogate-assisted passive design optimization applications

#### 4.1.1. Shape and geometry optimization

Seven studies demonstrated the capabilities of the S-BPO by optimizing only the shape and geometry parameters under diverse climatic conditions. For example, Ref. [62] optimized residential energy use intensity (EUI) and thermal comfort in a humid continental climate, whereas Ref. [70] sought to minimize residential heating and cooling loads in a Mediterranean climate. These studies promptly acquired preliminary design solutions by exploring a wide range of geometrical building configurations, orientations, or glazing areas. Similarly, in a humid continental climate, modifications to the interior atriums in a commercial building, as proposed in [71], achieved a reasonable reduction in the total, cooling, and lighting EUI.

The same group included studies that investigated SM-assisted urban form design optimization, which drastically reduced the time required for microclimate performance evaluations. For example, by altering the dimensions and spacing of an urban canyon in a subtropical climate [72], asserted the effectiveness of combining quadratic regression SMs and NSGA-II to improve pedestrian thermal comfort and wind velocity. In addition, by altering the locations and heights of 12 high-rise buildings [73], combined ANN approximations and NSGA-II to optimize five indoor and outdoor microclimate performance indicators. The study acknowledged that low prediction accuracies owing to the large model scale and complexity could be refined by investigating other sample sizes and SM types.

#### 4.1.2. Comprehensive passive parameters optimization

Another group of twenty-eight studies explored the potential impact of a wider range of passive design parameters on optimization, including the thermophysical, radiational, and optical properties of the building envelope. Among these studies, one subgroup focused on optimizing the fenestration properties for improved energy and thermal performances, spanning various methods, glazing system designs, and climatic regions. For instance, Ref. [74] combined Linear Regression (LR) SMs with NMSM to optimize a highly glazed façade in a temperate oceanic climate. Additionally, Ref. [44] employed ANN SMs and Teaching-Learning-based Optimization (TLBO) to enhance the properties of climate-adaptive aerogel glazing in a humid subtropical climate, while Ref. [75] integrated GP SMs with NSGA-II optimization for a double glazing system in a humid continental climate. Despite the promising results obtained by these studies, they highlighted the potential effectiveness of optimizing additional building design components for future optimizations, as showcased in [76].

In this vein [77], effectively achieved optimal design configurations for improved LC environmental impacts or energy consumption by combining a variety of office envelope parameters in a warm summer Mediterranean climate. Similarly, in a humid continental climate [78], optimized the LC carbon emissions (LCCE) and LC cost (LCC) of a passively designed residential building, with further improvement in energy efficiency by allocating a hybrid ventilation strategy. Hybrid ventilation demonstrates the influence of passive design strategies on energy efficiency during specific seasons [79], as it operates based on the indoor thermal comfort acceptability limits of the adaptive comfort model in [80].This adaptive model was employed, for instance, in Ref. [81] to define thermal comfort ranges in a sports hall in a humid continental climate during the transition seasons. The optimization of 14 passive design variables significantly improved the thermal energy consumption and comfort via night ventilation and an efficient AC system. Additionally [82], highlighted the role of the adaptive comfort model in unraveling the contradiction between thermal comfort and energy reduction targets, justifying the exclusion of thermal comfort optimization objectives when employing natural or hybrid ventilation modes.

Other studies have effectively employed daylight and visual comfort optimization, achieving satisfactory results while considerably accelerating complex optimization processes. For example, in a dense urban district with a temperate humid climate [83], combined ANN SMs with Self-adaptive Differential Evolution algorithm (jDE) to optimize each façade of five vertically aligned zones along a high-rise building. Furthermore, in the hot-summer Mediterranean climate, the study was developed in [63,84] by comprising ten stacked zones (along 60 floors), two façade design scenarios, and additional design variables, such as floor height and zone orientation. The intended design resulted in the development of 20 ANN SMs with a large number of design variables, which clearly outperformed the traditional optimization method of applying the same parameter values to the entire building. Nonetheless, the studies remarked on the benefits and suitability of integrating daylighting objectives with other BP indicators, such as energy demand and thermal comfort in [51], or EUI and envelope cost in [22].

#### 4.1.3. Integrating passive and active systems parameters

Twenty-five S-BPO studies incorporated optimization parameters related to active systems, such as the heating or cooling system type, set points, or airflow. For example, in a tropical climatic region [85], demonstrated a tradeoff between the conflicting objectives of energy consumption and thermal comfort in an educational building. The window area and wall thickness were optimized with eight HVAC system parameters, resulting in higher energy consumption for the entire optimal set of solutions. However, the selected optimal design substantially reduced the thermal comfort, meeting an acceptable range while moderately increasing the energy consumption. Another comprehensive study [86] developed an optimization framework for a nearly zero educational building in a humid subtropical climate by modifying ten distinct parameters. This study demonstrated significant BP improvements by simultaneously investigating the interactions and tradeoffs between annual thermal comfort and LC (50 years) objectives for energy consumption, AC energy consumption, carbon emissions, and cost.

In retrofit optimization studies [87], combined ANN-based SMs with MOGA to achieve cost-optimal retrofit solutions for office buildings in a Mediterranean climate. In this study, various performance indicators were improved by introducing 12 retrofit variables, including envelope materials, boiler and chiller types, and roof area for photovoltaic (PV) panels. A similar intention to minimize the financial burden of retrofitting in a humid continental climate was demonstrated in [56], utilizing a dataset of 10,368 residential retrofit scenarios for the ANN approximation and GA. While the optimal solutions revealed lower GHG emissions by increasing investment costs, the tradeoff favored retrofit decisions by prioritizing the LC economic benefits. Using similar methods in a temperate oceanic climate [88], optimized LC carbon emissions and financial savings by retrofitting a building stock of 95,500 dwellings. The expansive bottom-up approach necessitates the development of a second layer of surrogates based on the S-BPO of a representative sample of houses, providing less accurate but rapid retrofitting assessments.

### 4.2. Sensitivity of key design variables in distinct regions

Identifying the most influential design factors and their relationships with building performance can reduce the complexity and time required for the S-BPO. Therefore, twenty-one studies, covering various locations and building types incorporated sensitivity analysis (SA) as a preliminary phase of their optimization processes, focusing on their influence on building performance improvement or SM accuracy. In a monsoon-influenced hot-summer humid continental climate, Ref. [26] performed a global SA in a high-rise residential building by considering the effects of passive design parameters on energy demand, LC environmental impact, and LCC. The number of optimizable variables was reduced from 22 to 12, identifying airtightness, number of occupants (excluded from optimization), WWR, SHGC, and U-values as the most influential parameters. Similarly [89], conducted a sequential SA on the heating energy demand for a rural house, promoting glazing and envelope material parameters, South-facing WWR, and airtightness over other passive design parameters such as building orientation and dimensional proportions. Moreover [78], demonstrated that window material types, overhang depth, and WWR in the north and south façades of a residential building provided the maximum impact on LCCE and LCC compared with the envelope insulation thickness, building orientation, and WWRs of the other façades.

For other building types within the same region, Ref. [58] evaluated the influence of 28 design parameters on carbon emissions, thermal comfort, and LCC in an office building, averaging the outcomes of 11 SA methods. Sixteen key parameters were selected based on their combined contribution to three objectives, including setpoints, infiltration rate, wall and roof U-values, and orientation. A similar arrangement of key parameters was adopted in a subsequent LC optimization study [60], which identified fan and motor efficiency and supply air temperatures as the least impactful parameters for all objectives. Correspondingly, in a generally cooler climate of a warm summer humid continental region, Ref. [90] demonstrated the transcendence of envelope parameters, such as U-values and WWRs, in affecting the energy performance of a school building.

In a humid subtropical climate, Ref. [59] reduced the optimization complexity of energy consumption, thermal comfort, and daylight illuminance in an educational building by omitting the least influential variables, such as wall density and specific heat. A similar exclusion can be found in a more complex case study by [91], emphasizing the significance of thermostat setpoints, envelope thermal properties, WWR, and certain window types. Similarly, by comparing the optimal solutions of various subsets of 35 design and operational parameters [61], verified the predominance of AC systems and building envelope variables for improved residential energy consumption. However, this study demonstrated markedly superior solutions by considering all parameters simultaneously, even when providing minimal impact on any objective.

In tropical savanna climates, the building energy performance is mostly affected by the radiational properties of the external surfaces. A single-variable analysis in an office building by Ref. [92] showed that the cooling energy load and thermal discomfort increased with an increase in the window area, whereas increasing the number of horizontal louvers positively impacted all objectives, including UDI. In an educational building in another area with the same climate, the most influential passive design parameters on energy consumption were investigated by [93], which included the envelope's SHGC values, window areas, orientation, and window U-value. By contrast, the wall U-value and specific heat were among the least influential parameters, along with the ground slab properties.

In a hot-summer Mediterranean climate [94], adopted the best subset regression method for seven variables to predict electricity consumption, natural gas consumption, and thermal comfort in an office building. Each SM exhibited certain inputs that included a heating setpoint and ventilation and infiltration rates, while omitting the heating water temperature and roof U-value owing to their minimal impact on prediction accuracy. In the same climate [95], calibrated a residential thermal and energy model for retrofit optimization, including an SA of 58 uncertain parameters. In this study, the solar factor, infiltration rate, thermal conductivity, and thickness of the external envelope are among the most impactful parameters on energy performance.

In a hot semi-arid climate [96], investigated the relative importance of 11 carefully selected design parameters for residential building optimization. This study confirmed the superiority of wall and roof U-values, infiltration rate, and glazing material to most impact energy performance, while including floor U-value among the impactful parameters on thermal comfort. Alternatively, a residential S-BPO study in a hot, arid climate [69] demonstrated the significant impact of insulation thickness on cooling energy consumption and carbon emissions, with the cooling setpoint being the key parameter for thermal comfort.

Overall, the SA conducted in these studies proved to be significant for improving the efficiency and accuracy of S-BPO, emphasizing the importance of tailoring optimization strategies to specific building contexts and types. However, generalized conclusions for each performance indicator per climate necessitate a comprehensive investigation of consistent variable types and ranges. A comprehensive study was conducted by Ref. [79], integrating SA with S-BPO of passive design parameters on residential energy demand in five locations of distinct climates. The study adjusted various modeling and optimization techniques by excluding the least influential variables for each location. It is worth mentioning that adopting specific model settings and variable ranges can lead to the insignificance of widely used parameters, such as the infiltration rate in a humid subtropical climate [97] or thermostat setpoints in a temperate oceanic climate [98,99].

### 4.3. Adapting surrogate-assisted optimization techniques to diverse climatic conditions

Most S-BPO studies have been based on the approximated prediction of building performance under specific meteorological conditions. Recognizing the variability of solutions across different building contexts [63], applying the same approximation methods can lead to inaccurate predictions and potentially misguided optimal solutions. In response, 13 studies explored the capabilities of the S-BPO to accommodate varying climatic conditions in distinct regions or due to climate change effects, as summarized below.

#### 4.3.1. Climatic regions

Four studies addressed national (or local) climatic variations, as demonstrated in [100], combining polynomial regression SMs with NMSM, and [57], combining MFNN SMs with MOPSO. Both studies optimized residential thermal performance, including thermal energy, across six locations within three adjacent regions with hot, arid, semi-arid, and hot-summer Mediterranean climates. However, they revealed diverse passive design recommendations based on location, even for minor climatic variations within the same region. With similar objectives [101], achieved cost-optimal passive retrofit solutions for two educational building designs in three locations of Mediterranean climates by integrating ninety-six MFNN SMs and NSGA-II optimization. These studies suggest the development of separate SMs with sufficient sample sizes to accurately predict specific performance indicators for each location. This was demonstrated in five additional studies that investigated the S-BPOs in widely differentiated climatic regions, often employing SMs with location-specific structures and configurations. This variability can involve different input parameters, as demonstrated in [79], or varying the number of hidden neurons in MFNN-based SMs, as shown in [52].

Overall, replicating the same optimization process across various locations initially requires an exceptional computational budget. Nevertheless, appropriate techniques to reduce the number of simulated samples or the model structure complexity can facilitate the speed of convergence of climate-specific optimal solutions.

#### *4.3.2. Long-term climate change effects*

Only four studies addressed the influence of future climate change (CC) scenarios on optimal design solutions, a consideration valued by most of the reviewed studies to improve performance throughout a building's lifespan. In a temperate oceanic climate, Ref. [49] optimized LC performance, including cost and carbon reduction, for two university buildings by integrating passive, active, and renewable retrofit measures. This study utilized actual energy and weather datasets for one year to predict the future energy consumption based on HadCM3 climate projections. Additionally, in a humid subtropical climate [102], considered two future CC scenarios for each decade until 2099 to optimize the energy demand, winter thermal comfort, and UDI of an educational building by altering the passive design parameters. Both the S-BPO studies derived optimal design solutions for multiple climate scenarios, highlighting the ineffectiveness of historical weather data. Accordingly [103], conducted an uncertainty-based S-BPO study for a zero-carbon office building in a similar climate by relying on decades of historical and future weather data. This study emphasized the significant impact of uncertain design inputs, particularly weather conditions, on the selection of optimal solutions.

### *4.4. Recent advancements of surrogate-assisted optimization methods*

#### *4.4.1. Software interoperability*

The initial stage of S-BPO, which involves generating a performance database within the parameter space, requires a combination of sampling, modeling, and experimental tools. The transition between software tools for this and subsequent stages, including SM development, optimization, and post-processing, often promotes interoperability challenges. Several studies addressed these concerns by incorporating methods that facilitate seamless transitions or provide feedback interfaces, particularly for designers with limited coding skills. For example, visual programming environments, such as Dynamo in [90] or Grasshopper (GH) in [62] or [22], can streamline the entire S-BPO workflow by representing the design space, developing SMs, running optimization loops, and visualizing results. These studies attested that utilizing a single software platform supported by embedded modules and Python scripts enhances architects' capabilities to find optimal solutions, eliminating manual data conversion across various platforms.

In contrast, Integrated Development Environments (IDEs) for languages such as R, MATLAB, and Python are frequently utilized for surrogate modeling, even after employing a visual programming platform, to enhance flexibility in training options. In addition, in cases such as [104], separately executing the SM development process can provide better support for cloud-based computing via a web application, enabling flexible adjustments to the final model parameters based on the designer's preferences. Flexible design environments in similar studies have shown the effectiveness of SMs in providing a comprehensive view of the design space for exploring various non-performance-related solutions. Examples of such environments were demonstrated in [65] and [64], which further enhanced the interoperability by combining the existing functionality of visual programming environments with the ability to attain instantaneous performance feedback and higher customization control over the design space, thereby reducing the need for text-based coding.

#### *4.4.2. Adaptive sampling*

The static sampling methods adopted in the reviewed studies primarily emphasized the distribution of sample points within the design space, often without considering feedback from the simulated results. In contrast, adaptive sampling is a sequential space-filling technique that relies on this feedback, effectively balancing the exploration and exploitation of the design space [105]. Exploration aims to progressively cover under-sampled regions, while exploitation leverages the simulated data to add new sample points to complex, non-linear regions, potentially identifying optimal solutions [36]. Garud et al. [106] reviewed various adaptive sampling methods that can be applied throughout the entire design space. Among these, the LOLA-Voronoi algorithm, which incorporates local linear approximation (LOLA), was recently applied in BP prediction [105], demonstrating sample efficiency for accurate SMs by potentially maximizing the information gain with each new sample.

Incorporating an optimization algorithm to sampling-infill is a more nuanced approach that was demonstrated, albeit theoretically, in eight of the reviewed studies. For example, RBFN-based SM and NSGA-II were utilized in [107] to optimize residential energy and thermal performance and in [67] to improve the energy consumption and construction cost of office buildings, demonstrating the potential for iteratively refining rough SMs in each instance of the optimization process. Additionally, Ref. [48] progressively refined GP interpolation (kriging) models by adding new samples during the LC MOO using NSGA-II, highlighting computational efficiency, especially when considering climate change effects.

Although GP and RBFN-based SMs are typically well-suited for adaptive sampling due to their flexibility in guiding more accurate predictions, other SM types have been recommended for handling a higher number of design parameters. For instance, Ref. [52] effectively managed 15 design and operational variables by using progressively refined MFNN SMs, developed over 40 iterations of Ant Colony Optimization Algorithm for Continuous Domains (ACOR). In each iteration, new MFNNs were tuned using a 10-fold grid search cross validation and trained based on incrementally adding 50 new promising samples identified from the previous iteration. Despite the demonstrated effectiveness, especially for low sample sizes, the complexity of this framework may limit its broader adoption for future research. Alternatively, the surrogate-based optimization algorithm of RBFOpt [108] can be employed to iteratively adjust the Radial Basis Function (RBF) properties, providing sufficiently accurate approximations with potentially less complexity. This approach has been made accessible through GH [109], empowering designers to explore the adaptive sampling benefits of optimizing various model scales within a single platform. However, extensive benchmarking studies, as in ref [110], demonstrate the potential of such surrogate-based algorithms to become trapped in local optima, despite their performance at smaller function evaluation budgets, highlighting the need to test a wide variety of algorithms for specific design problems [111].

#### *4.4.3. Deep learning methods*

DL models use multiple layers to learn complicated features in a hierarchical manner, gradually forming more abstract representations of the raw data. The hierarchical feature learning process enables variations of such models to perform exceptionally well on complex tasks such as image recognition, language processing, and sequence modeling [112]. Recently, various forms of DL models were employed for time-series building performance prediction, such as MFDNNs in [113], deep Recurrent Neural Networks (RNNs) with Long Short-Term Memory (LSTM) layers in [114], and Convolutional Neural Networks (CNNs) combined with LSTM in [115]. These advanced forms of DL models have been effectively incorporated into various optimization and sensitivity analysis tasks, as reviewed by [116]. The optimization typically involves combining DL models with innovative BP prediction techniques to enhance the operational and management strategies of active systems, as in [117,118], or renewable energy systems, as in [119]. These innovative models can capture and utilize the intricate patterns in performance data, thereby providing a significant potential for the field of building design optimization. However, the applicability and reliability of these models are yet to be well explored, except for MFDNNs that mostly predict aggregated annual values of BP.

Considering the structure of typical DL models, it is worth mentioning that the effectiveness of a deep structure primarily depends on its suitability for the prediction problem [120]. Four hidden layers with nine neurons were sufficient to accurately predict the optimal start moment of the setback temperature, with an $R^2 \approx 1$ [121]. In addition, Ref. [62] demonstrated that exceeding three layers for predicting EUI and four layers for predicting thermal comfort can even increase the error values. While these studies opted for a moderately deep architecture, complex prediction problems may benefit from an even deeper structure. For instance, after comparing the performance of different DNN settings using GA optimization, Ref. [122] selected nine or ten hidden layers to accurately predict hourly energy consumption from time signature and weather profile features. In contrast, a later study, Ref. [49], employed the same hybrid GA-NN method to demonstrate that a single hidden layer with 140–200 neurons could accurately reveal the comprehensive patterns between similar inputs and the annual and monthly heating and electrical energy demand. A similar shallow structure with a high number of neurons achieved the highest accuracy, as demonstrated in the comparative studies by Refs. [90,55,123], and [124], highlighting the importance of evaluating various architectures based on the specific prediction task.

#### 4.4.4. *Ensemble learning and hybrid methods*

Twelve studies employed ensemble methods to mitigate SM accuracy limitations by aggregating their outputs, which showcased various techniques that provided a comprehensive understanding of their BP prediction and optimization applications. For example, Ref. [125] adopted bootstrap aggregation (bagging) to enhance the energy prediction accuracy of a relatively simple learning algorithm of a decision tree regressor (DTR). Alternatively, boosting ensemble techniques, such as the Gradient Boosting Machine (GBM) in [126] or Gradient Boosted Regression Trees (GBRT) in [70], progressively enhance the model performance by adjusting the errors of sequentially trained base learners. Although infrequently adopted in comparable studies, these model types were selected to predict residential thermal loads because of their accuracy and flexibility.

Another advanced form of the ensemble method was exemplified in [60], integrating the LC performance predictions of optimized SMs based on RF, GBRT, and ANN. Similarly, in [43], the Genetic Aggregation Response Surface (GARS) enhanced the Computational Fluid Dynamics (CFD) approximation by generating a population of SMs with optimized weight coefficients, incorporating polynomial regression and kriging. Moreover, in [90], the hybrid model of Group Last Squares-SVM (GLS-SVM), combining Group Method of Data Handling (GMDH) network with Last Squares-SVM, was promoted among the 11 SMs for S-BPO because it attained the highest accuracy in predicting energy and thermal performance.

Furthermore, ensemble methods based on the same algorithm were adopted, for instance, in [52], demonstrating the superiority of combining five ANN-based SMs with different random weight initializations to improve the S-BPO performance over accurate single models. Likewise, Ref. [127] developed an ensemble of ten Deep Feedforward ANNs (DFANNs) of four hidden layers and 280 nodes per layer of modified architectures based on slightly different datasets. The averaged residential energy demand and internal heat gain outputs of the ensemble approach neutralized any over- or underestimation of the base models, achieving values much closer to the actual performance.

On the other hand, a hybrid model involves integrating of two or more AI techniques to enhance the overall performance by leveraging their combined strengths. Optimizing the model structure and hyperparameters using an optimization algorithm is a form of model hybridization that can assist in minimizing the simulated sample size while maintaining reliable accuracy levels. For example, Ref. [57] demonstrated the effectiveness of the adopted S-BPO approach using only thirty-five simulations by incorporating PSO to optimize the weights and biases of MFNN-based SMs with 11 inputs. However, the overall performance was attributed to the entire methodology, including the careful sample selection based on trial-and-error evaluations and the development of separate models for each output per location. To address these issues [55], proposed a framework to generate and optimize an ANN-based SM with 13 inputs to simultaneously predict residential exergy destruction, thermal comfort, and LCC. Even though only 100 simulated samples were systematically selected via LHS, representing 2.2e-8 % of the entire design space, the extensive GA optimization, over 2,000 iterations of ANN structures and hyperparameters, could achieve an overall $R^2$ of 0.95.

#### 4.4.5. *Generalized surrogate models*

A generalized SM incorporates representations of varied contexts in its training data, allowing it to handle diverse scenarios without the need for retraining. This approach enhances efficiency and scalability of SMs for broader applications, as they can accommodate different climates, building types, or architectural forms. An earlier generalization attempt for various climates, as demonstrated by [128], featured aggregated abstractions of weather data, in terms of cooling and heating degree days, to estimate energy consumption across five climate zones in China. Alternatively, properly trained generalized SMs are expected to handle inputs nearly as effectively as physics-based simulations, including detailed weather features. This was demonstrated by the location-independent SM developed by Westermann et al. [129] (https://buildingenergy.ninja/), which estimates heating and cooling load profiles for any location using the learned features from annual multivariate weather data. The original model [130] leverages deep temporal convolutional neural networks (TCN) to combine learned features from hourly weather profiles with building design features, achieving an $R^2$ of 0.9971 in predicting annual heating loads in unseen locations. Even though the shallow MFNN with a selected set of weather features yielded comparable performance with an $R^2$ of 0.9951, using heating degree days (HDD) alone drastically reduced the model's accuracy. This underscores the unparalleled significance of feature selection, particularly for expanded search spaces.

In addition to climate generalization, SMs can ensure robust performance by accommodating various architectural configurations and functional parameters. One approach is component-based modeling (CBM), where separate SMs are trained according to the decomposition level of a complex simulation model. A high-level model decomposition was introduced by Li et al. [131], which divides architectural forms into multiple simple blocks to train the neural networks, thereby ensuring the SMs' adaptability to various architectural forms and sizes. On the other hand, a nuanced decomposition level was introduced by Geyer and Singaravel [132], which develops an SM to predict the heat flow per construction component, such as walls, windows, and floors, for each thermal zone. The outcomes of these models are aggregated to finally predict annual heating and cooling load. Besides generalizability to other building forms, CBM can even offer design granularity by easily identifying and developing design solutions tailored to a critical zone [133]. In addition, since CBM can be time-intensive, advanced training methods for improved efficiency, such as transfer learning in [134], or multi-task learning (MTL) DNN in [133] were developed and evaluated over various design cases. These advancements provide the basis for further investigation in future BPO studies.

#### 4.4.6. Advanced optimization techniques

Few studies have employed recently developed algorithms in BPO, demonstrating their potential to handle complex, high-dimensional problems while providing more accurate solutions. This includes, for example, the Two-Archive Evolutionary Algorithm for Constrained MOO (C-TAEA) for the multi-objective optimization problem (MOP) in [60], and jEDE and Covariance Matrix Adaptation with Evolution Strategy (CMA-ES) for the single-objective optimization problem (SOP) in [84]. These algorithms have demonstrated superior performance in several related studies when compared to widely adopted algorithms. However, selecting the best-performing optimization algorithm is case-specific because simpler direct search methods, such as NMSM in [99,100], can be sufficiently effective.

This thereby underscores the crucial impact of comparing the performance of various optimization algorithms, particularly when the true optimal set of solutions is unknown or prohibitively expensive to determine. In response, thirteen studies have incorporated criteria to benchmark algorithms based on their effectiveness, convergence rate, and exploration behavior. The comparisons showed that, despite their predominance in this review, GA variants, such as NSGA-II, outperformed other algorithms in only four cases. For example, Multi-Objective Harmony Search algorithm (MOHS) provided higher coverage rate and hypervolume than NSGA-II when optimizing investment cost and energy consumption in Ref. [135]. Similarly, the results from Multi-Objective Ant Lion Algorithm (MOALO) and MOPSO were retrieved due to their higher-quality solutions in Refs. [61] and [96], respectively, compared to other EAs.

Moreover, advancements in optimization techniques involved post-processing the optimal solutions and finding tradeoffs among conflicting objectives. For example, the best equilibrium solution proposed by [60] is a straightforward method for achieving a suitable balance among LCCE, LCC, and thermal comfort using simple formulas on the Pareto front and base case values. Another approach to decision-making was employed, for instance, between energy consumption, carbon emissions, and thermal comfort by Ref. [69], or between energy demand, LCC, and environmental LCA by Ref. [26], using Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS). Anyway, broadening the scope of experiments to include various decision-making techniques is advisable, as a comparison of the Linear Programming Technique for Multidimensional Analysis of Preference (LINMAP), TOPSIS, and Shannon's Entropy by Ref. [72] revealed that Shannon's Entropy provided the minimum deviation from the ideal solution.

### 4.5. Limitations and future research

The context-specific nature of S-BPO makes it improbable to standardize a universal framework for the entire process, thereby potentially promoting unique considerations for realizing problem-specific trade-offs between prediction accuracy, computational efficiency, and design performance quality. Accounting for these typical concerns, the key limitations of the SM development and optimization processes are discussed below, offering insights into directions for future research.

- **Limited prediction capabilities:** Although most developed SMs are sufficiently accurate within their training data limits, effectively encompassing diverse contexts such as building designs, types, or locations remains questionable. To improve the generalization capabilities across contexts, certain studies from the review sample habitually imposed either preparing separate simulated datasets for each variation, such as the distinct climates in [57], or developing a huge dataset covering possible combinations, as demonstrated in [127] and [126]. Synthesizing these data from scratch is computationally demanding and may become infeasible for globally diversified contexts. A potential solution to both challenges lies in the creation and global sharing of comprehensive building performance data. Another approach involves incorporating highly generalized SMs supported by sophisticated forms of DNNs.

  With the recent advancements of CBM, future work could explore combining these models with climate generalization techniques, even using simplified weather representations. For example, one of the location-independent SMs in [130] utilized engineered weather features (manually selected) such as HDDs and location coordinates, along with the mean and standard deviation of dry-bulb temperature, relative humidity, and global horizontal solar radiation over a year. To further streamline the climate generalization of already complex component-based models, simpler weather features could be employed while maintaining similar accuracy. One possible solution is the approach by Staffell at al. [136] (https://www.demand.ninja/), which calculates HDDs and CDDs based on Building Adjusted Internal Temperature (BAIT), accounting for the effects of solar irradiance, humidity, and wind speeds on building's energy consumption. Consequently, a single SM input feature may effectively represent the variations in the most impactful weather variables, potentially providing accurate predictions while maintaining model manageability for S-BPO incorporation.
- **Inaccuracy compared to real-world data:** BP simulation models can prompt a discordance drawback between the actual and simulated performances, often resulting from dissimilar assumptions, such as climate, equipment usage, and occupant behavior, from real-world conditions. Intensive reliance on simulated data would accumulate, or even exacerbate, these uncertainties in the approximated predictions of SMs. For example, Ref. [124] predicted the actual energy performance of urban blocks using an SM trained from a simulation-based optimization, revealing a

predominant overestimation despite proper model validation. Therefore, prediction accuracy can be enhanced by incorporating actual building performance data, taking into account the unpredictable factors in SM development. However, limited data availability may result in certain inadequacies, as demonstrated in [45], highlighting the inability to generalize the S-BPO findings of improved energy and comfort performance for office buildings owing to data constraints. Nevertheless, this study demonstrates the significant impact of occupant behavior variables on improving prediction accuracy.

Another perspective involves fully calibrated energy models, as exemplified in a review by Osman and Ouf [137] that focused on residential energy demand. These models rely on extensive datasets obtained from large-scale energy monitoring and surveys to generate realistic occupancy patterns. Although incorporating their outputs into SM development may enhance the effectiveness of the S-BPO, data scarcity in certain locations remains a shortcoming in creating such models. Furthermore, excessive information from detailed energy use profiles is often more relevant for energy management and the optimization of operational-related variables. Even when relying on generic occupant settings for simulations, a thorough validation of synthetic data is crucial to ensure consistency with actual performance.

Furthermore, S-BPO can significantly expedite the process of quantifying and potentially reducing inherent uncertainties on BP evaluations, enabling the exploration of diverse input scenarios that would be computationally prohibitive with traditional simulations. This was demonstrated, for instance, in the uncertainty-based S-BPO study by [26], by considering the utility cost and HVAC efficiency uncertainties, leading to a drastic shift in the optimal solutions. Overall, a performance-oriented building design supported by SMs is expected to provide more robust optimization methodologies [48] by accommodating real-world input variations.

- **Underutilization of sophisticated optimization algorithms:** Comparative studies of various algorithms often favored newly adopted variations, such as NSGA-III in [58] or CMA-ES in [84]. These variants usually introduce improved mechanisms for handling many objectives or navigating complex problems more efficiently and effectively. Taking the jEDE algorithm as an example, its adaptive parameter control and ensemble mutation strategies significantly improved convergence rate and solution quality, outperforming other optimization algorithms in 19 out of 20 benchmark problems [138]. Despite the unparalleled performance of jEDE, its applications in BPO are still limited to SOPs, leading to the continued reliance on less efficient or outdated optimization techniques to address more intricate challenges. The delay in adopting similar advanced methods in BPO can further widen the knowledge gap between optimization practices in BPO and those in other fields. For instance, recently developed DE variants are already capable of handling MOPs effectively, as demonstrated in [139] and [140], often outperforming hybrid and newly developed algorithms. Additionally, self-adaptation mechanisms can be applied to other MOO algorithms, such as the distribution index in the polynomial mutation operator of NSGA-II [141], significantly improving the effectiveness of an already well-established algorithm. The adaptation of advanced algorithms can be facilitated by leveraging and continuously updating Python-based libraries, such as Mealpy [142], or Geatpy [143], potentially offering robust frameworks for integrating self-adaptive and hybrid optimization algorithms with SMs trained within a unified IDE.
- **Arbitrary selection of optimization methodologies:** It is widely recognized that the effectiveness of an optimization algorithm is problem-specific, as no algorithm can comprehensively outperform others in all problems [144]. However, nearly 82 % of the reviewed studies selected the BPO algorithm arbitrary, without assessing the comparative performance of alternative techniques. In fact, the findings from relevant comparison studies frequently contradict the prevailing choices, imposing the critical need for performing such rigorous pre-experimentations. While the comparisons can include recently developed algorithms and hybrid approaches, expanding these comparisons to include surrogate-based optimization methods, i.e., by incorporating adaptive sampling techniques, could further refine the selection process. Additionally, it is crucial to recognize the significant computational efficiency provided by S-BPO, which facilitates more comprehensive evaluations by incorporating various static parameters and initial settings for each algorithm. This approach, coupled with standardized evaluation metrics, can provide the basis of a nuanced methodology tailored to any problem-specific context.
- **Misalignment in SM selection:** Effective S-BPO necessitates the careful selection of sufficiently accurate SMs, which is typically achieved by employing a suitable sampling strategy or by comparing various model types and settings. Nearly all studies have employed error metrics from test set predictions, such as $R^2$, RMSE, or MAPE, for this process. These metrics are widely used to assess the overall accuracy of SMs throughout the entire design space, ensuring their suitability for various applications, such as early deign exploration or sensitivity analysis. However, overreliance on these global metrics might obscure poor performance in critical regions where optimization algorithms explore more intensively, eventually leading to suboptimal or incorrect solutions. To overcome this fatal methodological flaw, it's crucial to complement these global error metrics with additional validation methods to ensure the model's robustness. One method that have been recently applied in the field of BP prediction is the uncertainty-aware building energy SMs [145], which utilize Bayesian methods to quantify uncertainty in BP approximation. By identifying points with the largest prediction errors, these models enable a hybridization process where the surrogate model is then supplemented with additional simulations, potentially enhancing overall accuracy. Additionally, these models can be beneficial for adaptive sampling, assisting the optimization algorithms in refining the rough SM during the initial iterations.
- **Lack of a robust adaptive sampling technique:** Globally accurate SMs do not necessarily guarantee trustworthy optimal solutions, a challenge that certain studies have addressed by increasing the sample size, as shown in [68,69], to enhance proximity to simulation-based optimal solutions. However, increasing the sample size without incorporating simulated feedback is both ineffective and inefficient because it can lead to an oversampling in non-critical regions while leaving complex areas unexplored. Alternatively,

progressively refining the approximated solution space by means of adaptive sampling seem more suitable for BPO applications. However, only a few studies have utilized this approach, often with several limitations in their methodologies. For example, the adaptive sampling approach in [52] performed exceptionally well only during the initial iterations, compared to a simulation-based optimization using the same ACOR algorithm, as adding more than 1000 samples failed to improve the model's performance. In addition, excessive SM refinement further accumulates the optimization time, either by requiring new simulations or retraining complex SM with new data points, making this technique less suitable for large-scale case studies, as suggested by [84]. Hence, it is crucial to find a balance between the efficiency and quality of the optimal solutions. This balance requires the exploration of diverse aspects of the sampling strategy, while incorporating more advanced adaptive sampling techniques for enhanced efficiency. For instance, the Smart Sampling Algorithm (SSA) by Garud et al. [146], has been extensively evaluated against other methods by Ref. [147], demonstrating promising efficiency improvements.

Finally, the extraction of the approximate optimal solutions is not always the ultimate step. Beyond verification with high-fidelity simulations, the S-BPO requires further manual refinement to reach reliable design conclusions. This necessitates extensive approximated evaluations to encompass a broader solution range, potentially enhancing designers' perceptions of non-performance-related aspects of the optimization problem [47].

## 5. Conclusions

This review thoroughly investigates seventy-two recent studies that have predominantly incorporated ML-based SMs into the field of BPO, emphasizing the impact of passive design strategies on environmental sustainability across diverse contexts. The performance evaluation accuracy is particularized for extended scrutiny, revealing attainable high prediction efficacy coupled with considerable computational efficiency potential. This accentuates the S-BPO's capability to address previously intractable design problems, potentially achieving significant sustainability improvements by incorporating an unprecedented dimensionality level of design variables.

Nonetheless, high-dimensional inputs should be simplified through SA, omitting less influential parameters from the optimization to reduce the computational burden. Consequently, decisive design parameters exhibit variations based on predefined design criteria such as internal gains and design constraints and contextual variations such as building types and climatic conditions. Additionally, S-BPO can encompass regional and temporal climatic variations, potentially generating diverse design solutions for the same base model tailored to the requirements of each climate. Although these models often involve computationally intensive sample generation for each variation, highly generalized SMs can extrapolate building performance across uncharted contexts, indicating promising pathways for future S-BPO research.

Incorporating innovative SM development techniques, such as model hybridization, ensemble learning, and advanced deep learning, is recommended to enhance the prediction accuracy with fewer training databases. Such training methods can assist in developing a single SM to predict combined, and sometimes contradictory, performance objectives. Furthermore, notable progress has been made in design-space exploration using unified software platforms. However, considerable work remains in developing fully functional and universally recognized *flexible design environments* that seamlessly integrate designer preferences into optimal solutions.

Future work should also consider the potential limitations that can pose challenges in achieving precise and reliable optimal solutions because blindly relying on SMs may result in misleading outcomes. Establishing rigorous strategies for SM evaluation and adaptive sampling for optimization purposes is essential to mitigate these limitations. In addition, the remarkable disparity in S-BPO techniques, which may follow contextual and design requirement variations, can further amplify existing limitations. It is noteworthy that context-specific factors, such as architectural design, climate, and even the selection of objective functions and design variables, play a significant role in determining the type and configuration of SMs and optimization algorithms. Therefore, even with the congruence demonstrated in most recent studies toward generally recognized ideal techniques and configurations, local experimentation with each stage of the S-BPO is still recommended to attain the most suitable, case-specific optimization workflow.

Ultimately, this review aims to guide future research by focusing on innovative methodologies for effective and efficient S-BPO applications across architectural, regional, and temporal variations. However, the reliability of these methodologies is influenced by the unexplored technical aspects of the review sample, which are pertinent to surrogate modeling and optimization algorithms. Hence, studies that compile and benchmark such algorithms, irrespective of their field, are integral to a holistic understanding of S-BPO research.

## CRediT authorship contribution statement

**Ibrahim Elwy:** Writing – original draft, Methodology, Data curation, Conceptualization. **Aya Hagishima:** Writing – review & editing, Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Appendix

**Table A1**
Summary of the reviewed studies, including case study, modeling, and optimization properties. Note: For comparison, the selected algorithms and sample sizes are indicated in **Bold**.

| # | Ref | Author(s) | Year | Building Type | Design stage *1 | Climate: Multi/CC *2 | Climate: Location (Climate *3) | SA *4 | Design parameters *5: UC | BG | FG | EM | FM | HV | Re | Ap | L | Surrogate model(s) *6 | Sampling strategy: Simulation tool(s) *7 | Sample size | Sample division (%) *8: Tr | V | Te | Sampling method | Check *9 | A/S *10 | Hyperparameter and structure optimization | LC *11 | OFs / SM Outputs *12: RE | EC | CE | TC | DD | VC | C | Ot | Performance optimization strategy: Algorithm(s) *1 | Population / Iterations | Tool(s) - Software |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | [107] | Chen, Y. and Shi, X. | 2023 | Residential | ED | - | -, northeast China | - | o | o | o | o | o | - | - | - | - | RBFN | EnergyPlus | 500 | 90 | | 10 | Latin square sampling | - | A | None | - | - | o | - | o | - | - | - | - | NSGA-II | 100, max iterations are 1000 | |
| 2 | [60] | Chen, R. et al. | 2023 | Office | ED | - | Qingdao, China (Dwa) | o | - | o | o | o | o | o | - | o | o | ELM (GBRT +RF+ANN) | EnergyPlus (LB&HB) | 16320 (after SA) | 70 | | 30 | FAST-**Sobol**-RS-QRS-LHS (by SimLab) | o | S | Grid Search, Random Search | o | - | - | o | o | - | - | o | - | **C-TAEA**, NSGA-II, NSGA-III | | Python platform |
| 3 | [90] | Hosamo, H. et al. | 2022 | Educational | ED | - | Kjevik, Kristiansand, Norway (Dfb) | o | - | o | o | o | o | o | - | o | o | Compare 11 ML algorithms (**GLS-SVM** was chosen) | IDA Indoor Climate and Energy (IDA ICE) | 8000 (16 days to sim.) | 80 | 10 | 10 | pairwise test | - | S | Grid Search | - | - | o | - | o | - | - | - | - | NSGA-II | 40, 400 generations max | Dynamo software (Optimo) |
| 4 | [69] | Saryazdi, S. et al. | 2022 | Residential | ED | - | Kuwait (BWh) | o | o | - | - | o | o | o | - | o | - | MFNN | jEPlus + EnergyPlus (DesignBuilder) | 9 datasets, 300 - 26000 (**9900**) | 80 | | 20 | LHS | o | S | Trial-and-error | - | - | o | o | o | - | - | - | - | MOGA | | Matlab |
| 5 | [43] | Kaseb, Z. and Montazeri, H. | 2022 | High-rise (no type) | ED | - | Rotterdam, the Netherlands (Cfb) | - | - | o | - | - | - | - | - | - | - | Compare: RSM, KG, NN, SVR, **GARS** | 3D steady RANS CFD, ANSYS/Fluent 19.3 | 211 (chosen to be **184** samples) | 88 | | 12 | 5 methods, including 3 Space-filling design, LHS, and Central Composite Design | o | S | None | - | o | - | - | - | - | - | - | - | GA | 100, 100 iteration max. | Ansys DesignXplorer |
| 6 | [44] | Zhou, Y. | 2022 | no-type | N/A | - | Guangzhou, China (Cfa) | - | - | - | o | - | o | - | - | - | - | ANN | Mathematical model | from 18 to **72** | 80 | | 20 | cross-entropy algorithm | o | S | Trial-and-error | - | - | - | - | o | - | - | - | o | TLBO | | |
| 7 | [55] | Kerdan, I. and Gálvez, D. | 2022 | Residential | ED | - | Baja California Sur, Mexico (BWh) | - | - | o | - | o | o | o | o | - | o | BPNN (DNN) | EnergyPlus + ExRET-Opt (Developed by Authors) | 100 | 95 | | 5 | LHS | - | S | GA | o | - | o | - | o | - | - | o | - | NSGA-II | 100, 100 generations max. | ANNEXE, Python-based software |
| 8 | [62] | Guo, H. et al. | 2022 | Residential | ED | - | Jinan, China (Dwa) | - | - | o | o | - | - | - | - | - | - | ANN (EC) - SVM (TC) | EnergyPlus (LB&HB) | 500 | 80 | | 20 | LHS | - | S | Trial-and-error | - | - | o | - | o | - | - | - | - | SPEA-2 | 86 iterations | GH, octopus |
| 9 | [86] | Wu, X. et al. | 2022 | Educational | N/S | - | Wuhan, China (Cfa) | - | - | - | o | o | o | o | - | - | - | RF | BIM (Revit) - EnergyPlus (DesignBuilder) | 88 | 80 | | 20 | Orthogonal experimental design | - | S | Grid Search | o | - | o | o | o | - | - | o | - | NSGA-III | 100 | |
| 10 | [53] | Zhou, Y. et al. | 2022 | Public (Library) | ED | - | Urumqi, China (BSk) | o | - | o | o | - | - | - | - | - | - | LightGBM | Radiance (LB&HB) / simple GH calculations | 5000 | 80 | | 20 | after iterative genetic optimization process (Wallacei) | - | S | Grid Search | - | - | - | - | - | o | - | - | o | NSGA-II | 50, 100 generations | GH, Wallacei |
| 11 | [58] | Chen, R. et al. | 2022 | Office | ED | - | Qingdao, China (Dwa) | o | - | o | o | o | o | o | - | o | o | BPNN | jEPlus - EnergyPlus + LB (for TC) | 13323 | 70 | | 30 | 3 methods: LHS-Sobol-FAST | o | S | Grid Search, Random Search | o | - | - | o | o | - | - | o | - | **NSGA-III** (compared with MOEA/D and NSGA-II) | 496, 50 iterations | PYMOO toolkit of Python |
| 12 | [91] | Xu, Y. et al. | 2022 | Educational | ED | - | Nanjing, China (Cfa) | o | - | o | o | o | o | o | - | - | - | MFNN | EnergyPlus | up to 3000 for each of 2 cases (skylihgt / no skylight) | 80 | | 20 | LHS | - | S | GA | - | - | o | - | o | o | - | - | - | Comparing: MOGA, **NSGA-II** and MOPSO | 50, 100 iterations | |
| 13 | [57] | Chegari, B. et al. | 2022 | Residential | both | Multi | 6 locations in Morocco (Csa, BSh, BWh) | - | - | o | o | o | o | - | - | - | - | MFNN | TRNSYS | 35 | 70 | 15 | 15 | trial-and-error-based evaluation simulations | - | S | PSO | - | - | o | - | o | - | - | - | - | MOPSO | 50, 70 iterations | Matlab |
| 14 | [49] | Luo, X. and Oyedele, L. | 2022 | Educational / Office | RE | CC | Bristol, UK (Cfb) | - | - | - | - | o | - | o | o | - | - | ANN | (real energy consumption data from bills) | | | | | | - | S | GA | o | - | o | o | - | - | - | - | - | PSO | | |
| 15 | [88] | Hey, J. et al. | 2023 | Residential | RE | - | Nottingham, UK (Cfb) | - | - | - | - | o | o | o | - | - | - | ANN | EnergyPlus | 40,000 | 70 | 15 | 15 | | - | A | Grid Search | o | - | o | o | - | - | - | o | - | GA | | |
| 16 | [92] | Zhang, J. and JI, L. | 2022 | Office | ED | - | Sanya, China (Aw) | o | - | o | o | - | - | - | - | - | - | BPNN | EnergyPlus + Radiance (LB&HB) | 1157 | 90 | | 10 | obtained from the 51 optimization generations | - | S | None | - | - | o | - | o | o | - | - | - | | 20, reached stable state after 51 generations | Octopus (GH) |
| 17 | [93] | Rajput, T. and Thomas, A. | 2022 | Educational | ED | - | Mumbai, India (Aw) | o | - | o | o | o | o | - | - | - | - | ANN | jEPlus + EnergyPlus (DesignBuilder) | 3000 | 70 | 15 | 15 | LHS | - | S | Trial-and-error | - | - | o | - | o | - | - | - | - | NSGA-II, then PSO for comparison | 200, 1000 generations | Matlab |
| 18 | [148] | Yang Y. et al. | 2022 | Residential | ED | - | N/S, HSCW area in China | - | - | - | - | o | - | - | - | - | - | BPNN | BECS | 800 | 70 | | 30 | | - | S | Trial-and-error | - | - | o | - | o | - | - | - | - | | | Matlab |
| 19 | [71] | Yuan, Y. et al. | 2022 | Commercial | ED | - | Beijing, China (Dwa) | o | - | o | - | - | - | - | - | - | - | MFNN | EnergyPlus (GH/Archsim) | 1439 | 70 | 10 | 20 | LHS | - | S | Trial-and-error | - | - | o | - | - | - | - | - | - | PSO | 50, 50 generations max | |
| 20 | [78] | Xue, Q. et al. | 2022 | Residential | ED | - | Harbin, China (Dwa) | o | - | o | o | o | o | - | - | - | - | ANN | EnergyPlus | 1000 | 70 | 15 | 15 | LHS | - | S | Trial-and-error | o | - | - | o | - | - | - | o | - | NSGA-II | 200, 100 generations max | |
| 21 | [81] | Yue, N. et al. | 2021 | Public (Sport) | N/S | - | Qingdao, China (Dwa) | - | - | o | o | o | o | - | - | - | o | Compare: **ANN-MLP**, RF, SVM, RBF | EnergyPlus + Eppy | 1600 | | | | LHS | o | S | None | - | - | o | - | o | - | - | - | - | NSGA-II | | |
| 22 | [59] | Xu, Y. et al. | 2021 | Educational | ED | - | Nanjing, China (Cfa) | o | - | o | o | o | o | o | - | - | - | ANN | EnergyPlus | 2000 | 80 | | 20 | Monte Carlo simulation | o | S | Grid Search | - | - | o | - | o | o | - | - | - | NSGA-II | 50, 100 generations max | |
| 23 | [124] | Wang, W. et al. | 2021 | Residential | ED | - | Jianhu, China (Cfa) | - | o | o | - | - | - | - | - | - | - | LSTM network | EPlus (HB) + LB (for solar hours/rad.) + DF (for UHI morphed EPW) | 1447 | 70 | 30 | | generated from the optimization process (Wallacei) | - | S | Trial-and-error | - | o | o | - | - | o | - | - | - | NSGA-II | 33, 100 iterations | GH, Wallacei X |
| 24 | [102] | Zou, Y. et al. | 2021 | Educational | ED | CC | Guangzhou, China (Cfa) | - | - | o | o | o | o | - | - | - | - | ANN | EnergyPlus (LB&HB) | 50000+ | 70 | 10 | 20 | Random (Colibri tool of GH) | - | S | Trial-and-error | - | - | o | - | o | o | - | - | - | NSGA-II | 400, 50 generations | Geatpy on Python |
| 25 | [45] | Amasyali, K. and El-Gohary, N. | 2021 | Office | N/A | - | Philadelphia, Pa, USA (Cfa) | - | - | - | o | - | - | o | - | o | o | Compared 6 ML alg. (**SVR** for EC, **KNN** for TC, and **SVM** for VC) | (from real-data sensors for 3 months) | | | | | | o | S | Grid Search | - | - | o | - | o | - | o | - | - | NSGA-II | 50, 700 generations | MATLAB |
| 26 | [73] | Wang, S. et al. | 2021 | Residential | ED | - | Beijing, China (Dwa) | - | o | o | - | - | - | - | - | - | - | ANN | GH-LB plugins, for UTCI: Eddy3D (OpenFOAM, Radiance) | Ranged from **200** to 50 | 70 | 15 | 15 | Dynamic Parameter Control System | - | S | Trial-and-error | - | - | - | - | o | o | o | - | o | NSGA-II | 150, 200 generations | MATLAB |
| 27 | [51] | Zou, Y. et al. | 2021 | Educational | ED | - | Guangzhou, China (Cfa) | - | - | o | o | o | o | - | - | - | - | ANN | EnergyPlus (LB&HB) | 50000+ | | | | Random (Colibri tool of GH) | - | S | Trial-and-error | - | - | o | - | o | o | - | - | - | NSGA-II | 400, 50 generations | Geatpy on Python |
| 28 | [26] | Jung, Y. et al. | 2021 | Residential | ED | - | Seoul, South Korea (Dwa) | o | - | o | o | o | o | - | - | - | - | Kriging | TRNSYS | 4000 | 90 | 10 | | LHS | - | S | Trial-and-error | o | - | o | o | - | - | - | o | - | NSGA-II | 200, 1200 generations | |
| 29 | [94] | Ghaderian, M. and Veysi, F. | 2021 | Office | RE | - | Kermanshah, Iran (Csa) | o | - | - | o | o | - | o | - | - | - | regression models | EnergyPlus | | | | | Box-Behnken design | o | S | N/A | - | - | o | - | o | - | - | - | - | NSGA-II | 50 | MATLAB |
| 30 | [84] | Ekici, B. et al. | 2021 | Office | ED | - | Izmir, Turkiye (Csa) | - | - | o | o | - | o | - | - | - | - | BPNN | DIVA for GH (took more than 55 days) | 20,000 (**1000** for each zone) | 80 | | 20 | LHS | - | S | Grid Search | - | - | - | - | - | o | - | - | - | RBFOpt, CMA-ES and **jEDE** | 10,000 function evaluations | GH (Opossum, Optimus) |
| 31 | [63] | Ekici, B. et al. | 2021 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

(continued on next page)

**Table A1** (*continued*)

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 32 | [127] | Pittarello, M. et al. | 2021 | Residential | ED | - | Venice, Italy (Cfa) | - | - | o | o | o | o | o | - | o | o | DFANN | EnergyPlus (one zone per building) | 600,000 | 75 | | 25 | | - | S | Trial-and-error | - | - | o | o | o | - | o | - | o | | | |
| 33 | [149] | Lin, Y. and Yang, W. | 2021 | Residential | ED | Multi | 5 locations in China (Dwa, Cfa, Cwb) | - | - | o | o | o | o | - | - | - | - | ANN | EnergyPlus (DesignBuilder) | 90 to **280** | 90 | 10 | | LHS (3 sizes) | o | S | None | - | - | o | - | o | - | - | - | - | exhaustive listing technique | | |
| 34 | [96] | Chegari, B. et al. | 2021 | Residential | both | - | Marrakech, Morocco (BSh) | o | - | o | o | o | o | - | - | - | - | MFNN | TRNSYS | 35 | 70 | 15 | 15 | Trial-and-error-based evaluation simulations | - | S | PSO | - | - | o | - | o | - | - | - | - | NSGA-II, **MOPSO** and MOGA | ranges between 25 and 100, both for pop. size & iterations | MATLAB |
| 35 | [46] | Sonta, A. et al. | 2021 | Office | N/A | - | Redwood city, CA, USA (Csb) | - | - | o | - | - | - | - | - | - | - | Compare: MLR, **RF**, SVR and ANN | (from real-data sensors for 132 full days) | | 80 | | 20 | | - | S | Grid Search | - | - | o | - | - | - | - | - | - | clustering-based optimization and GA | 100 generations | |
| 36 | [104] | Zhao, S. | 2021 | Office | ED | - | Hangzhou, China (Cfa) | - | - | o | o | - | - | - | - | - | - | ANN | EnergyPlus + Daysim (LB&HB) | 650 | 80 | | 20 | LHS | - | S | Trial-and-error | - | - | o | - | - | o | - | - | - | NSGA-II | | Geatpy on Python |
| 37 | [61] | Lin, Y. et al. | 2021 | Residential | ED | - | Shanghai, China (Cfa) | - | - | o | o | o | o | o | - | - | - | Compare: SLR, **BPNN**, SVM, RF | Radiance + EnergyPlus (DesignBuilder) | 800 | 90 | 10 | | LHS | - | S | Trial-and-error | - | - | o | - | - | - | o | - | - | Comparing: MOEA/D, NSGA-II, NSGA-III, MOPSO, MODA, **MOALO** | | |
| 38 | [99] | Hawila, A. and Merabtine, A. | 2021 | Educational | ED | - | Troyes, France (Cfb) | o | o | - | o | - | o | o | - | - | - | regression models | Modelica | 16 | | | | full factorial design | - | S | N/A | - | - | o | - | o | - | - | - | - | NMSM | | |
| 39 | [95] | Azevedo, L. et al. | 2019 | Residential | RE | - | Lisbon, Portugal (Csa) | o | - | - | - | o | o | o | - | - | o | ANN-MLP | EnergyPlus | 2200 | 91 | 9 | | LHS | - | S | Trial-and-error | - | - | o | - | o | - | - | o | - | GA | 20, 50, 10, 200 and 300 (2000 gen max) | |
| 40 | [56] | Zhang, H. et al. | 2022 | Residential | RE | - | British Columbia, Canada (-) | - | - | - | o | o | - | o | - | - | - | ANN-MLP | HOT2000 and HTAP | 10,368 | 70 | 20 | 10 | | - | S | GA | o | - | o | o | - | - | - | o | - | GA | | Python |
| 41 | [52] | Bamdad, K. et al. | 2020 | Office | ED | Multi | 2 locations in Australia (Cfb, Cfa) | - | - | o | o | o | - | o | - | - | - | ANN-MLP | EnergyPlus | 2,000 (maximum) | | | | initial sampling: LHS | o | A | Grid Search | - | - | o | - | - | - | - | - | - | ACOR, PSOIW (for simulation based) | | MATLAB / GenOpt |
| 42 | [64] | Brown, N. et al. | 2020 | Public (Sport) | ED | - | Boston, MA, USA (Dfa) | - | - | o | o | - | - | - | - | - | - | Compare: **RF** and Ensemble NN | Karamba + Diva (GH) | 1000 (for energy model), 12,000 (for structural model) | 72 | | 28 | LHS | - | S | Trial-and-error | - | - | o | - | - | - | - | - | o | NSGA-II | | |
| 43 | [65] | Brown, N. | 2020 | Public (Sport) | ED | - | Boston, MA, USA (Dfa) | - | - | o | o | - | - | - | - | - | - | Compare: ANN, KRR, **RF** | Karamba + Diva (GH) | | 96 | | 4 | LHS | - | S | None | - | - | o | - | - | - | - | - | o | GA | | GH (Galapagos) |
| 44 | [89] | Lee, B. et al. | 2020 | Residential | ED | - | Seoul, South Korea (Dwa) | o | - | o | o | o | o | - | - | - | - | ANN | PHPP | 10,000 | | | | random sampling (Mode Frontier) | - | S | None | - | - | o | - | - | - | - | - | - | sequential RSM | | |
| 45 | [22] | Sun, C. et al. | 2020 | Public (Library) | ED | - | Changchun, China (Dwa) | - | - | o | o | o | o | - | - | - | - | ANN | EnergyPlus + Radiance (LB&HB) | 200 | 60 | 40 | | LHS | - | S | GA | - | - | o | - | - | o | - | o | - | GA | 100, 50 generations | GH (Octopus) |
| 46 | [74] | Hawila, A. et al. | 2020 | Educational | ED | - | Troyes, France (Cfb) | o | o | - | o | - | o | - | - | - | - | regression models | Modelica | 32 | | | | two-level full factorial design | - | S | N/A | - | - | o | - | o | - | - | - | - | NMSM | | |
| 47 | [111] | Pereira, I. et al. | 2020 | no-type | N/S | - | Lisbon, Portugal (Csa) | - | - | - | o | o | - | - | - | - | - | Compare: RF and GPR | Radiance (for DD) | initially: 75 | | | | | - | A | None | - | - | - | - | - | o | - | o | - | Compare: SPEA2, NSGA-II, SMPSO and OMOPSO | | |
| 48 | [125] | Boghetti, R. et al. | 2020 | Residential | ED | - | Turin, Italy (Cfa) | - | o | - | - | - | - | - | - | - | - | bagging (to DTR) | (using real hourly consumption data) | | | | | | - | S | None | - | - | o | - | - | - | - | - | - | | | |
| 49 | [126] | Yigit, S. | 2021 | Residential | ED | Multi | 3 locations in Turkiye (Csa, BSk) | - | - | o | o | o | - | - | - | - | - | GBM | EnergyPlus | 30,000 | 80 | | 20 | LHS | - | S | Grid Search | o | - | o | - | - | - | - | - | - | NSGA-II | 100, 100 generations | DEAP (Python) |
| 50 | [68] | Bre, F. et al. | 2020 | Residential | ED | - | Parana, Argentina (Cfa) | - | - | o | o | o | o | - | - | - | - | MFNN | EnergyPlus | 1,600 | 91 | 9 | | LHS | o | S | None | - | - | o | - | o | - | - | - | - | NSGA-II | 64, 100 generations | DEAP (Python) |
| 51 | [83] | Ekici, B. et al. | 2019 | Office | N/S | - | N/S, temperate-humid climate | - | - | - | o | - | o | - | - | - | - | BPNN | DIVA (GH) | 2500 total (**500** per zone) | | | | Random (uniform distribution function, coded in C#) | - | S | Trial-and-error | - | - | - | - | - | o | - | - | - | self-adaptive differential evolution (jDE) | 30, 500 generations | C# |
| 52 | [103] | Li, H. et al. | 2019 | Office | ED | CC | Hong Kong (Cwa) | o | - | o | o | o | o | - | - | - | - | MFNN | jEplus - EnergyPlus | 212 | 85 | | 15 | LHS | - | S | Grid Search | - | - | o | - | o | - | - | - | - | GA/NSGA-II | 100 | Matlab |
| 53 | [85] | Nasruddin et al. | 2019 | Educational | N/S | - | West Java, Indonesia (-) | - | - | - | o | o | - | o | - | - | - | ANN-MLP | IESVE software | 250 | 60 | 40 | | LHS | o | S | Trial-and-error | - | - | o | - | o | - | - | - | - | MOGA | | |
| 54 | [98] | Hawila, A. et al. | 2019 | Educational | ED | - | Troyes, France (Cfb) | o | o | - | o | - | o | o | - | - | - | regression models | Modelica (Dymola) | 68 | 58 | | 42 | two-level full factorial design | - | S | None | - | - | - | - | o | - | - | - | - | NMSM | | |
| 55 | [150] | Orosa, J. et al. | 2019 | Office | RE | - | Galicia, Spain (Csb) | - | - | - | - | o | - | - | - | - | - | GRNN | (Real outdoor weather data and indoor conditions) | 300 per office and per season | 75 | 25 | | | - | S | None | - | - | - | - | o | - | - | - | - | | | |
| 56 | [151] | Si, B. et al. | 2019 | Public (tourist center) | ED | - | Nanjing, China (Cfa) | - | - | o | - | o | o | o | - | - | - | ANN | EnergyPlus | 2000 | 70 | 15 | 15 | LHS | - | S | Trial-and-error | - | - | o | - | o | - | - | - | - | **NSGA-II**, MOPSO, MOSA & ES | 40, 45 generations | modeFRONTIER, MATLAB |
| 57 | [72] | Du, Y. et al. | 2019 | Urban Canyons | ED | - | Hong Kong (Cwa) | - | o | o | - | - | - | - | - | - | - | Compared $1^{st}$ order and **$2^{nd}$ order regression** models | CFD Fluent (RANS Turbulence model) | 32 | 78 | 22 | | Box-Behnken Design (BBD) | - | S | N/A | - | - | - | - | o | - | - | - | o | NSGA-II | 200, 500 generations | |
| 58 | [152] | Aijazi, A. et al. | 2019 | Residential | RE | - | Lisbon, Portugal (Csa) | - | - | - | o | o | o | - | - | - | o | Kriging | EnergyPlus | | | | | LHS | - | S | None | - | - | o | - | - | - | - | o | - | GA | | MATLAB |
| 59 | [35] | Prada, A. et al. | 2018 | Residential | N/S | - | Milan, Italy (Cfa) | - | - | - | - | o | o | o | - | - | - | Comparing 8 models | TRNSYS | from 32 to 4096 | | | | | o | A | Trial-and-error | - | - | o | - | - | - | - | o | - | NSGA-II | | MATLAB |
| 60 | [79] | Chen, X. and Yang, H. | 2018 | Residential | ED | Multi | 5 locations in China (Dwa, Cfa, Cwb, Cwa) | o | - | o | o | o | o | - | - | - | - | SVM (MLR for SA) | EnergyPlus | 5610 | | | | Monte Carlo (LHS) | - | S | Random Search | - | - | o | - | - | - | - | - | - | NSGA-II | 20, 100 generations | R environment |
| 61 | [76] | Lin, Y. et al. | 2018 | Residential | N/S | - | Wuhan, China (Cfa) | - | - | - | o | o | - | - | - | - | - | MLR and ANN | EnergyPlus (DesignBuilder) | 450 | 90 | 10 | | LHS | - | S | None | - | - | o | - | o | - | - | - | - | GA | 29 runs (MLR GA), 14 runs (ANN-GA) | |
| 62 | [70] | Papadopoulos, S. et al. | 2018 | Residential | ED | - | Athens, Greece (Csa) | - | o | o | o | - | - | - | - | - | - | GBRT | Ecotect | 768 | 80 | | 20 | | - | S | Grid Search | - | - | o | - | - | - | - | - | - | GA | 2000, 15 generations | |
| 63 | [48] | Fraisse, G. et al. | 2017 | Residential | ED | CC | Savoie, France (Cfb) | - | - | - | - | o | - | o | o | - | - | Kriging | TRNSYS + EQUER (for LCA) | | | | | LHS (initial DoE) + adaptive sequential design | - | A | None | o | - | o | o | o | - | - | o | o | NSGA-II | | MATLAB |
| 64 | [87] | Ascione, F. et al. | 2017 | Office | RE | - | Naples, Italy (Csa) | - | - | o | o | o | o | o | - | - | o | ANN-MLP | EnergyPlus | | 90 | | 10 | LHS | - | S | Trial-and-error | - | o | o | - | o | - | - | o | - | MOGA | 50, 300 generations | MATLAB |
| 65 | [75] | Kim, Y. and Park, C. | 2017 | Office | N/S | - | Seoul, South Korea (Dwa) | - | - | - | - | - | o | - | - | - | - | GP | EnergyPlus (OpenStudio) | 500 / glazing system | | | | LHS | - | S | Bayesian Optimization | - | - | o | - | o | - | - | - | - | NSGA-II | 600, 1000 generations | MATLAB |

(*continued on next page*)

**Table A1** (*continued*)

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 66 | [82] | Chen, X. and Yang, H. | 2017 | Residential | N/S | Multi | Hong Kong (Cwa), Los Angeles (BSk) | - | - | o | o | o | o | - | - | - | - | MLR, MARS, and SVM | EnergyPlus | 5610 | | | | Monte Carlo (LHS) | - | S | None | - | - | o | - | - | - | - | - | - | NSGA-II | 20, 100 generations | R environment |
| 67 | [101] | Almeida, R. and De Freitas, V. | 2016 | Educational | RE | Multi | 3 locations in Portugal (Csb, Csa) | - | - | - | o | o | o | - | - | - | - | MFNN | EnergyPlus (DesignBuilder) | 1920 (**160** for each ANN) | 94 | 6 | | LHS | - | S | None | o | - | o | - | o | - | - | o | - | NSGA-II | 100, 100 generations | MATLAB |
| 68 | [77] | Azari, R. et al. | 2016 | Office | N/S | - | Seattle, WA, USA (Csb) | - | - | - | o | o | o | - | - | - | - | ANN | eQuest + LCA calc. and Athena IE | 91 | 70 | 30 | | | - | S | GA | o | - | o | o | - | - | - | - | - | NSGA-II | 50, 500 generations | |
| 69 | [47] | Wortmann, T. et al. | 2015 | Public (Church) | N/A | - | Singapore (Af) | - | - | - | o | - | - | - | - | - | - | RBF | DIVA (GH) | | | | | | - | A | None | - | - | - | - | - | o | o | - | - | RBFOpt, GA | 25, 4 generations | |
| 70 | [100] | Romani, Z. et al. | 2015 | Residential | N/S | Multi | 6 locations in Morocco (Csa, BSh, BWh) | - | - | - | o | o | o | - | - | - | - | polynomial regression | TRNSYS | 15, 25, 100, 110 (each for a polynomial type) | 50 | 50 | | D-optimal design | - | S | N/A | - | - | o | - | - | - | - | - | - | NMSM | | |
| 71 | [67] | Brownlee, A. and Wright, J. | 2015 | Office | ED | Multi | Birmingham, UK (Cfb), Chicago, IL, USA (Dfa), San Francisco, CA, USA (Csb) | - | - | o | o | o | o | o | - | - | - | RBFN | EnergyPlus | 1000 | | | | random | - | A | Trial-and-error | - | - | o | - | - | - | - | o | - | NSGA-II | 20, max 5000 evaluation | |
| 72 | [153] | Geyer, P. and Schlüter, A. | 2014 | Office | RE | - | Zurich, Switzerland (Cfb) | - | - | o | o | o | o | o | - | - | - | RSM | EnergyPlus | 128 | | | | Coordinate-Exchange Algorithm | o | S | Trial-and-error, GA- NMSM | - | - | o | - | o | - | - | - | o | Design Space Exploration | | |

*[1] Design stages: Early design (ED), Retrofit (RE), Not applicable (N/A), Not specified (N/S). *[2] Multiple climatic regions (Multi), Accounting for Climate Change effects (CC).
*[3] Climatic region abbreviations for Köppen-Geiger climate classification are explained in Ref. [51] (Table 2). *[4] Studies that conducted sensitivity Analysis of input variables.
*[5] Design Parameters: Urban context (UC), Building Geometry or shape (BG), Fenestration Geometry (FG), Envelope Material (EM), Fenestration Material (FM), HVAC / system (HV), Renewables (Re), Appliances (Ap), and Lights (L).
*[6] RSM: Response Surface Methodology, KG: Kriging, SVR: Support Vector Regression, BPNN: Backpropagation NN, DNN: Deep NN, ANN-MLP: Multi-Layer Perceptron NN, LSTM: Long Short-Term Memory, KNN: K-Nearest Neighbors, MLR: Multiple Linear Regression, SLR: Simple Linear Regression, KRR: Kernel Ridge Regression, GPR: Gaussian Process Regression, GRNN: Generalized Regression NN, MARS: Multivariate Adaptive Regression Splines.
*[7] LB&HB: Ladybug and Honeybee plugins in Grasshopper (GH), BECS: Building Energy Conservation Software, PHPP: Passive House Planning Package, IESVE: Integrated Environmental Solutions Virtual Environment, Dymola: Dynamic Modeling Laboratory, Athena IE: Athena Impact Estimator for Buildings.
*[8] Sample division: Training (Tr), Validation (V), and Testing (Te). *[9] Check the effect of the sampling strategy on the accuracy of SMs.
*[10] Adaptive (A) or Static (S) sampling. *[11] Studies that conducted Lifecycle assessments for cost or environmental impacts.
*[12] Objective functions (OFs) or Surrogate Model (SM) outputs: Renewable Energy yield (RE), Energy Consumption / demand (EC), $CO_2$ Emissions (CE), Thermal Comfort (TC), Dynamic Daylight (DD), Visual Comfort (VC), Cost / energy cost (C), Others (Ot).
*[13]C-TAEA: Two-Archive Evolutionary Algorithm for Constrained multi-objective optimization, SPEA-2: Strength Pareto Evolutionary Algorithm 2, CMA-ES: Covariance Matrix Adaptation with Evolution Strategy, MODA: Multi-Objective Dragonfly Algorithm, MOALO: Multi-Objective Ant Lion Algorithm, ACOR: Ant Colony Optimization Algorithm for Continuous Domains, MOSA: Multi-Objective Simulated Annealing, ES: Evolution Strategy.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.enbuild.2024.114769.

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